

## True or false?

$(9 + 4) \times 2$  is 2 more than  $9 + 4$ .

**A.** True

**B.** False

Which expression is 4 times as large as  $125 - 25$ ?

**A.**  $(125 - 25) \div 4$

**B.**  $4 \times (125 - 25)$

**C.**  $125 - 25 \times 4$

**D.**  $4 + (125 - 25)$

Without calculating, how many times as large is  
 $10 \times (63 + 7)$  as  $63 + 7$ ?

Times as large: \_\_\_\_\_

SET 43 • 6.EE.A.2.b

# FORWARD 1

Combine like terms with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$$4x + 3 - 2x + 5$$

Simplified: \_\_\_\_\_

Simplify.

$$0.5y + 1.25y$$

Simplified: \_\_\_\_\_

Which pair shows like terms?

**A.**  $3a$  and  $-7a$

**B.**  $x^2$  and  $x$

**C.**  $3a$  and  $3b$

**D.**  $4x$  and  $4$

Simplify.

$$-3(x - 2) + 5x$$

Simplified: \_\_\_\_\_



## True or false?

$$\frac{1}{2}x + \frac{1}{4}x = \frac{3}{4}x$$

**A.** True

**B.** False

SET 43 • 6.EE.A.2.b

# FORWARD 2

Interpret the parts of a linear function • Grade 8 • Functions • 8.F.B.4

The cost to bowl is  $C = 4.5n + 12$ , where  $n$  is the number of games.

What does 12 represent?

Meaning: \_\_\_\_\_

In  $y = -3x + 7$ , what is the rate of change?

**A.**  $x$

**B.**  $3$

**C.**  $-3$

**D.**  $7$

The gas left in a car's tank is  $G = 14 - 0.04m$ , where  $m$  is miles driven.

What is the initial value?

Initial value: \_\_\_\_\_

A car's distance from home is  $d = 60t + 25$ , where  $t$  is hours.

What does 60 represent?

- A.** The speed in miles per hour
- B.** The starting distance
- C.** The total distance
- D.** The total time

## True or false?

In  $y = 8x$ , the initial value is 0.

**A.** True

**B.** False

# Evaluate expressions and formulas for given values

|                   |                                                   |                   |
|-------------------|---------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 6 main question                             | <b>6.EE.A.2.c</b> |
| <b>BACKWARD 1</b> | Whole-number exponents                            | <b>6.EE.A.1</b>   |
| <b>BACKWARD 2</b> | Order of operations with grouping symbols         | <b>5.OA.A.1</b>   |
| <b>BACKWARD 3</b> | A letter stands for a number                      | <b>3.OA.D.8</b>   |
| <b>BACKWARD 4</b> | Multiplying fractions                             | <b>5.NF.B.4.a</b> |
| <b>FORWARD 1</b>  | Evaluate expressions with rational numbers        | <b>7.NS.A.3</b>   |
| <b>FORWARD 2</b>  | Volume formulas for cylinders, cones, and spheres | <b>8.G.C.9</b>    |



SET 44 • 6.EE.A.2.c

# MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate  $3x^2 + 2$  when  $x = 4$ .

Value: \_\_\_\_\_

The volume of a cube is  $V = s^3$ .

Find  $V$  when  $s = \frac{2}{3}$  foot.

$V =$  \_\_\_\_\_

The perimeter of a rectangle is  $P = 2l + 2w$ .

Find  $P$  when  $l = 7.5$  cm and  $w = 4$  cm.

$P =$  \_\_\_\_\_

Evaluate  $2(a + b)$  when  $a = 3.5$  and  $b = 4$ .

**A.** 7.5

**B.** 9.5

**C.** 11.5

**D.** 15

Evaluate  $\frac{x}{4} + 5y$  when  $x = 12$  and  $y = 2$ .

Value: \_\_\_\_\_

SET 44 • 6.EE.A.2.c

# BACKWARD 1

Whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$4^2$$

Value: \_\_\_\_\_



Evaluate.

$$\left(\frac{1}{2}\right)^3$$

Value: \_\_\_\_\_

**True or false?**

$$3^2 = 6$$

**A.** True

**B.** False

$$2^5 = ?$$

**A.** 10

**B.** 16

**C.** 25

**D.** 32

Evaluate.

$$5^2 + 1$$

Value: \_\_\_\_\_

SET 44 • 6.EE.A.2.c

# BACKWARD 2

Order of operations with grouping symbols • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$3 \times (16 + 2)$$

Value: \_\_\_\_\_

Evaluate.

$$2 \times (6 + 4)$$

Value: \_\_\_\_\_

**True or false?**

$$(3 + 4) \times 2 = 11$$

**A.** True

**B.** False



Evaluate.

$$(12 \div 4) + 10$$

**A.** 1

**B.** 6

**C.** 13

**D.** 22

Evaluate.

$$[2 \times (5 + 3)] - 4$$

Value: \_\_\_\_\_

SET 44 • 6.EE.A.2.c

# BACKWARD 3

A letter stands for a number • Grade 3 • Operations & Algebraic Thinking •  
3.OA.D.8

If  $n = 6$ , what is  $n + 9$ ?

Value: \_\_\_\_\_

## True or false?

If  $a = 4$ , then  $a \times 5 = 20$ .

**A.** True

**B.** False

If  $k = 7$ , what is  $3 \times k$ ?

**A.** 10

**B.** 21

**C.** 37

**D.** 73

If  $m = 20$ , what is  $m - 8$ ?

Value: \_\_\_\_\_

If  $t = 3$ , what is  $4 \times t + 2$ ?

Value: \_\_\_\_\_



SET 44 • 6.EE.A.2.c

# BACKWARD 4

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{2}{3} \times \frac{2}{3} \times \frac{2}{3}$$

Product: \_\_\_\_\_

Multiply.

$$6 \times \frac{1}{3}$$

Product: \_\_\_\_\_

## True or false?

$$\frac{2}{3} \times \frac{2}{3} = \frac{4}{6}$$

**A.** True

**B.** False

$$\frac{3}{4} \times 8 = ?$$

**A.**  $\frac{3}{32}$

**B.** 2

**C.** 6

**D.** 24

Multiply.

$$\frac{1}{2} \times 12$$

Product: \_\_\_\_\_

SET 44 • 6.EE.A.2.c

# FORWARD 1

Evaluate expressions with rational numbers • Grade 7 • The Number System •  
7.NS.A.3

Use  $C = \frac{5}{9}(F - 32)$  to find  $C$  when  $F = -4$ .

$C =$  \_\_\_\_\_



Evaluate  $-2x + 5$  when  $x = -3.5$ .

Value: \_\_\_\_\_

Evaluate  $a^2 - b$  when  $a = -3$  and  $b = 4$ .

**A.**  $-13$

**B.**  $-5$

**C.**  $5$

**D.**  $13$

Evaluate  $\frac{(x+y)}{2}$  when  $x = -6.5$  and  $y = 2.5$ .

Value: \_\_\_\_\_

## True or false?

When  $x = -2$ , the value of  $x^3 + 1$  is  $-7$ .

**A.** True

**B.** False

SET 44 • 6.EE.A.2.c

# FORWARD 2

Volume formulas for cylinders, cones, and spheres • Grade 8 • Geometry •  
8.G.C.9

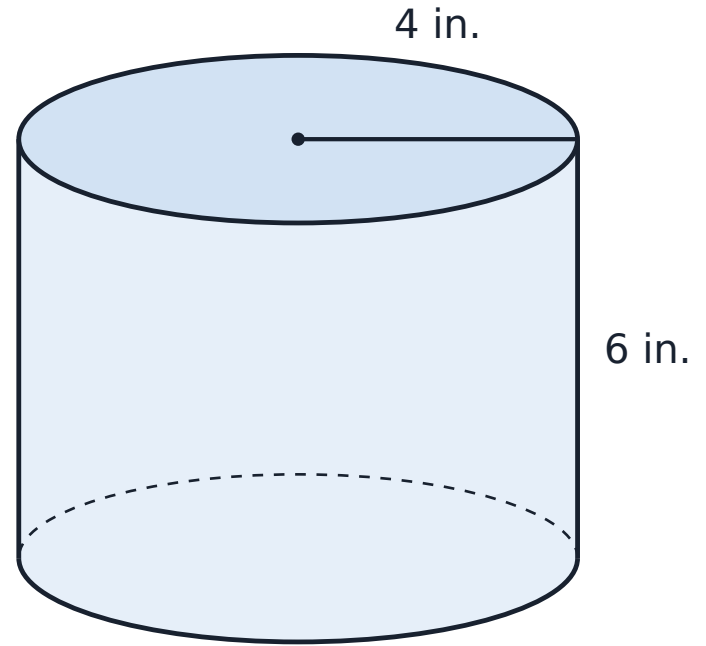
A cylinder has a radius of 3 cm and a height of 10 cm.  
Find its volume. Use 3.14 for  $\pi$ .

Volume: \_\_\_\_\_

Find the volume of the cylinder.

Use 3.14 for  $\pi$ .

Volume: \_\_\_\_\_



A cone has a radius of 3 units and a height of 4 units.

What is its volume in terms of  $\pi$ ?

**A.**  $36\pi$

**B.**  $48\pi$

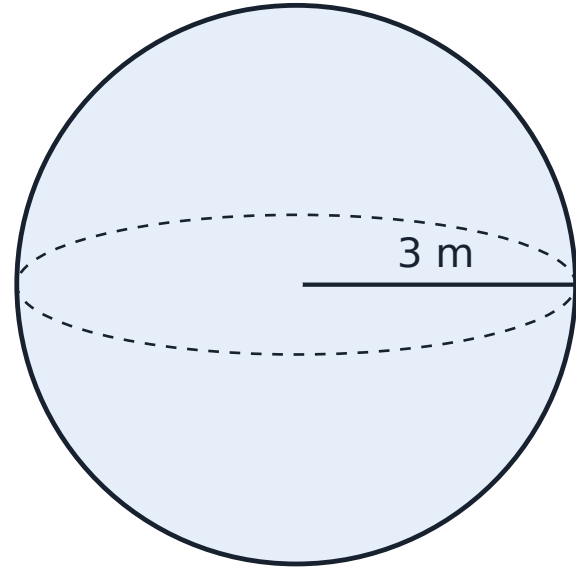
**C.**  $4\pi$

**D.**  $12\pi$



Find the volume of the sphere  
in terms of  $\pi$ .

Volume: \_\_\_\_\_



## True or false?

A cylinder with a radius of 2 and a height of 5 has a volume of  $20\pi$ .

**A.** True

**B.** False

# Use the distributive property to expand expressions

|                   |                                                        |                   |
|-------------------|--------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 6 main question                                  | <b>6.EE.A.3</b>   |
| <b>BACKWARD 1</b> | Distributive property with numbers                     | <b>3.OA.B.5</b>   |
| <b>BACKWARD 2</b> | Equal groups                                           | <b>3.OA.A.1</b>   |
| <b>BACKWARD 3</b> | Multiplying a decimal by a whole number                | <b>5.NBT.B.7</b>  |
| <b>FORWARD 1</b>  | Expand linear expressions with rational coefficients   | <b>7.EE.A.1</b>   |
| <b>FORWARD 2</b>  | Solve equations by expanding and collecting like terms | <b>8.EE.C.7.b</b> |

SET 45 • 6.EE.A.3

# MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.3

Expand.

$$3(4 + x)$$

Expression: \_\_\_\_\_

Expand.

$$5(3a - 2)$$

Expression: \_\_\_\_\_

Which expression is equivalent to  $7(2m + 5)$ ?

- A.**  $9m + 35$
- B.**  $2m + 35$
- C.**  $14m + 5$
- D.**  $14m + 35$

Expand.

$$2(x + 3y + 4)$$

Expression: \_\_\_\_\_



Expand.

$$4(0.5y + 3)$$

Expression: \_\_\_\_\_

SET 45 • 6.EE.A.3

# BACKWARD 1

Distributive property with numbers • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

**True or false?**

$$6 \times (4 + 3) = 6 \times 4 + 6 \times 3$$

**A.** True

**B.** False

## Fill in the blank.

$$5 \times (2 + 7) = 5 \times 2 + 5 \times \underline{\quad}$$

Blank: \_\_\_\_\_

Which is equal to  $3 \times 14$ ?

**A.**  $30 + 4 + 3$

**B.**  $3 \times 10 + 4$

**C.**  $3 \times 10 + 3 \times 4$

**D.**  $3 + 10 \times 4$

**True or false?**

$$7 \times 12 = 7 \times 10 + 2$$

**A.** True

**B.** False

Fill in the blank.

$$8 \times (5 + 1) = 8 \times 5 + \underline{\hspace{2cm}}$$

Blank: \_\_\_\_\_

SET 45 • 6.EE.A.3

# BACKWARD 2

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1



## True or false?

$3 \times (4 + 2)$  means 3 groups of  $(4 + 2)$ .

**A.** True

**B.** False

Which means 4 groups of  $(5 + 1)$ ?

**A.**  $4 + (5 + 1)$

**B.**  $4 \times (5 + 1)$

**C.**  $4 \times 5 + 1$

**D.**  $(4 + 5) \times 1$

How many groups of  $(2 + 6)$  are in  $5 \times (2 + 6)$ ?

Groups: \_\_\_\_\_

**True or false?**

$$2 \times (3 + 4) = (3 + 4) + (3 + 4)$$

**A.** True

**B.** False

Write  $6 \times 3$  as a sum of equal groups.

Sum: \_\_\_\_\_

SET 45 • 6.EE.A.3

# BACKWARD 3

Multiplying a decimal by a whole number • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

$$4 \times 0.5$$

Product: \_\_\_\_\_

Multiply.

$$3 \times 1.2$$

Product: \_\_\_\_\_



**True or false?**

$$6 \times 0.25 = 1.5$$

**A.** True

**B.** False

$$5 \times 0.4 = ?$$

**A.** 0.2

**B.** 2

**C.** 5.4

**D.** 20

Multiply.

$$8 \times 0.75$$

Product: \_\_\_\_\_

SET 45 • 6.EE.A.3

# FORWARD 1

Expand linear expressions with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Expand.

$$\frac{1}{2}(8x - 6)$$

Expression: \_\_\_\_\_

Expand.

$$-4(2y - 3)$$

Expression: \_\_\_\_\_

Which is equivalent to  $0.5(6a + 10)$ ?

- A.**  $3a + 5$
- B.**  $6a + 5$
- C.**  $3.5a + 5$
- D.**  $3a + 10$

Expand.

$$-\frac{2}{3}(9m + 3)$$

Expression: \_\_\_\_\_



**True or false?**

$$-2(x - 7) = -2x + 14$$

**A.** True

**B.** False

SET 45 • 6.EE.A.3

# FORWARD 2

Solve equations by expanding and collecting like terms • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$3(x + 2) = 5x - 4$$

x = \_\_\_\_\_

Solve for x.

$$2(4x - 1) - 3x = 18$$

x = \_\_\_\_\_

Solve for x.

$$-2(x - 5) = 3(x + 5)$$

**A.**     $-5$

**B.**     $-1$

**C.**     $1$

**D.**     $5$

Solve for x.

$$6(x - 1) = 4(x + 2)$$

x = \_\_\_\_\_

Solve for x.

$$5(2x + 1) = 3(3x + 4)$$

x = \_\_\_\_\_

# Use the distributive property to factor expressions

|                   |                                                            |                   |
|-------------------|------------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 6 main question                                      | <b>6.EE.A.3</b>   |
| <b>BACKWARD 1</b> | Greatest common factor                                     | <b>6.NS.B.4</b>   |
| <b>BACKWARD 2</b> | Distributive property with numbers                         | <b>3.OA.B.5</b>   |
| <b>BACKWARD 3</b> | Factors                                                    | <b>4.OA.B.4</b>   |
| <b>FORWARD 1</b>  | Factor linear expressions with rational coefficients       | <b>7.EE.A.1</b>   |
| <b>FORWARD 2</b>  | Use factoring to decide how many solutions an equation has | <b>8.EE.C.7.a</b> |



SET 46 • 6.EE.A.3

# MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.3

Factor using the greatest common factor.

$$24x + 16$$

Expression: \_\_\_\_\_

Factor using the greatest common factor.

$$18y + 30$$

Expression: \_\_\_\_\_

Which shows  $15a + 25b$  factored using the greatest common factor?

- A.**  $15(a + 10b)$
- B.**  $5(3a + 5b)$
- C.**  $3(5a + 25b)$
- D.**  $5(3a + 25b)$

Factor using the greatest common factor.

$$36m + 27n$$

Expression: \_\_\_\_\_

Factor using the greatest common factor.

$$14x + 21$$

Expression: \_\_\_\_\_

SET 46 • 6.EE.A.3

# BACKWARD 1

Greatest common factor • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 16.

GCF: \_\_\_\_\_



Find the greatest common factor of 18 and 30.

GCF: \_\_\_\_\_

## True or false?

The greatest common factor of 15 and 25 is 5.

- A. True
- B. False

What is the greatest common factor of 36 and 27?

- A.** 3
- B.** 6
- C.** 9
- D.** 12

Find the greatest common factor of 14 and 21.

GCF: \_\_\_\_\_

SET 46 • 6.EE.A.3

# BACKWARD 2

Distributive property with numbers • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

**True or false?**

$$8 \times 3 + 8 \times 2 = 8 \times (3 + 2)$$

**A.** True

**B.** False

Fill in the blank.

$$6 \times 4 + 6 \times 5 = 6 \times (4 + \underline{\quad})$$

Blank: \_\_\_\_\_

Which is equal to  $7 \times 2 + 7 \times 9$ ?

**A.**  $14 + 9$

**B.**  $7 \times 2 \times 9$

**C.**  $7 \times (2 + 9)$

**D.**  $7 + 2 + 9$



**True or false?**

$$5 \times 6 + 3 \times 6 = 6 \times (5 + 3)$$

**A.** True

**B.** False

Fill in the blank.

$$9 \times 7 + 9 \times 3 = \underline{\hspace{1cm}} \times (7 + 3)$$

Blank: \_\_\_\_\_

SET 46 • 6.EE.A.3

# BACKWARD 3

Factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factors of 16.

Factors: \_\_\_\_\_

**True or false?**

6 is a factor of 30.

**A.** True

**B.** False

Which number is a factor of 27?

- A.** 4
- B.** 6
- C.** 9
- D.** 12

List all the factors of 21.

Factors: \_\_\_\_\_

**True or false?**

8 is a factor of 36.

**A.** True

**B.** False



SET 46 • 6.EE.A.3

# FORWARD 1

Factor linear expressions with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Factor out 2.5.

$$2.5x + 10$$

Expression: \_\_\_\_\_

Factor out  $-4$ .

$$-4x + 12$$

Expression: \_\_\_\_\_

Which is equivalent to  $\frac{1}{2}x + 3$ ?

**A.**  $\frac{1}{2}(x + \frac{3}{2})$

**B.**  $\frac{1}{2}(x + 3)$

**C.**  $\frac{1}{2}(x + 6)$

**D.**  $2(x + 6)$

Factor out 0.6.

$$0.6y - 1.8$$

Expression: \_\_\_\_\_

**True or false?**

$$-5a - 15 = -5(a + 3)$$

**A.** True

**B.** False

SET 46 • 6.EE.A.3

# FORWARD 2

Use factoring to decide how many solutions an equation has • Grade 8 •  
Expressions & Equations • 8.EE.C.7.a

Factor the left side to decide. How many solutions does the equation have?

$$6x + 9 = 3(2x + 3)$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions



How many solutions does the equation have?

$$12x + 8 = 4(3x + 1)$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

How many solutions does the equation have?

$$10x - 15 = 5(2x - 3)$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

How many solutions does the equation have?

$$8x + 4 = 4(x + 1)$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

What number goes in the box so that the equation has infinitely many solutions?

$$14x + 21 = 7(2x + \square)$$

$\square =$  \_\_\_\_\_

# Combine like terms to write equivalent expressions

|                   |                                               |                   |
|-------------------|-----------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 6 main question                         | <b>6.EE.A.3</b>   |
| <b>BACKWARD 1</b> | Repeated addition as multiplication           | <b>3.OA.A.1</b>   |
| <b>BACKWARD 2</b> | Changing the order and grouping of addends    | <b>3.OA.B.5</b>   |
| <b>FORWARD 1</b>  | Combine like terms with rational coefficients | <b>7.EE.A.1</b>   |
| <b>FORWARD 2</b>  | Solve equations by collecting like terms      | <b>8.EE.C.7.b</b> |

SET 47 • 6.EE.A.3

# MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.3

Write an equivalent expression with one term.

$$y + y + y + y$$

Expression: \_\_\_\_\_

Simplify.

$$5x + 3x$$

Expression: \_\_\_\_\_



Which expression is equivalent to  $7a + 2 + 3a$ ?

**A.**  $7a + 5a$

**B.**  $10a + 5$

**C.**  $10a + 2$

**D.**  $12a$

Simplify.

$$4m + 6 + 2m + 1$$

Expression: \_\_\_\_\_

Simplify.

$$9n - 4n + 3$$

Expression: \_\_\_\_\_

SET 47 • 6.EE.A.3

# BACKWARD 1

Repeated addition as multiplication • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

**True or false?**

$$4 + 4 + 4 = 3 \times 4$$

**A.** True

**B.** False

Write  $7 + 7 + 7 + 7$  as a multiplication expression.

Expression: \_\_\_\_\_

What does  $4 \times 9$  mean?

- A.** 4 groups of 9
- B.** 9 minus 4
- C.** 4 more than 9
- D.** The number 49

**True or false?**

$$2 + 2 + 2 + 2 + 2 = 2 \times 2$$

**A.** True

**B.** False



Write  $9 + 9$  as a multiplication expression.

Expression: \_\_\_\_\_

SET 47 • 6.EE.A.3

# BACKWARD 2

Changing the order and grouping of addends • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

**True or false?**

$$4 + 6 + 2 = 4 + 2 + 6$$

**A.** True

**B.** False

Fill in the blank.

$$7 + 5 + 3 = 7 + 3 + \underline{\quad}$$

Blank: \_\_\_\_\_

Which is equal to  $(8 + 2) + 5$ ?

**A.**  $8 + 2 \times 5$

**B.**  $8 \times (2 + 5)$

**C.**  $(8 - 2) + 5$

**D.**  $8 + (2 + 5)$

**True or false?**

$$9 + 1 + 9 = 9 + 9 + 1$$

**A.** True

**B.** False

Fill in the blank.

$$6 + 3 + 4 = 6 + 4 + \underline{\hspace{1cm}}$$

Blank: \_\_\_\_\_

SET 47 • 6.EE.A.3

# FORWARD 1

Combine like terms with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1



Simplify.

$$2.25y + 0.75y$$

Expression: \_\_\_\_\_

Simplify.

$$\frac{1}{2}x + \frac{1}{4}x$$

Expression: \_\_\_\_\_

Simplify.

$$6x - 2 + 3x + 7$$

Expression: \_\_\_\_\_

Simplify.

$$-3a + 7a - 2$$

**A.**  $4a - 2$

**B.**  $4a + 2$

**C.**  $2a$

**D.**  $-10a - 2$

**True or false?**

$$2.5\text{m} - \text{m} = 1.5\text{m}$$

**A.** True

**B.** False

SET 47 • 6.EE.A.3

# FORWARD 2

Solve equations by collecting like terms • Grade 8 • Expressions & Equations •  
8.EE.C.7.b

Solve for x.

$$3x + 5x - 4 = 20$$

x = \_\_\_\_\_

Solve for  $y$ .

$$7y - 2y + 3 = 2y + 18$$

$y =$  \_\_\_\_\_



Solve for a.

$$4a + 6 - a = 2a + 10$$

**A.**  $-4$

**B.**  $\frac{16}{5}$

**C.**  $4$

**D.**  $16$

Solve for n.

$$0.5n + 1.5n - 3 = n + 5$$

n = \_\_\_\_\_

Solve for k.

$$6k - 2k + 1 = 3k - 4$$

k = \_\_\_\_\_

# Identify when two expressions are equivalent

|                   |                                                        |                   |
|-------------------|--------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 6 main question                                  | <b>6.EE.A.4</b>   |
| <b>BACKWARD 1</b> | Commutative and associative properties                 | <b>3.OA.B.5</b>   |
| <b>BACKWARD 2</b> | Evaluate an expression for a given value               | <b>6.EE.A.2.c</b> |
| <b>BACKWARD 3</b> | Equal groups                                           | <b>3.OA.A.1</b>   |
| <b>FORWARD 1</b>  | Rewrite expressions to show how quantities are related | <b>7.EE.A.2</b>   |
| <b>FORWARD 2</b>  | One, none, or infinitely many solutions                | <b>8.EE.C.7.a</b> |

SET 48 • 6.EE.A.4

# MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.4

## True or false?

$m + m + m + m$  and  $4m$  are equivalent expressions.

**A.** True

**B.** False

Which expression is equivalent to  $2(x + 4)$ ?

**A.**  $x + 8$

**B.**  $2x + 4$

**C.**  $2x + 6$

**D.**  $2x + 8$

## True or false?

$3(x + 2)$  and  $3x + 2$  are equivalent expressions.

**A.** True

**B.** False



Which expression is equivalent to  $6a + 4a$ ?

**A.**  $6a + 4$

**B.**  $10a$

**C.**  $24a$

**D.**  $10a^2$

Which pair of expressions is equivalent?

**A.**  $5(a + 3)$  and  $5a + 15$

**B.**  $5(a + 3)$  and  $a + 15$

**C.**  $5(a + 3)$  and  $5a + 3$

**D.**  $5(a + 3)$  and  $8a$

SET 48 • 6.EE.A.4

# BACKWARD 1

Commutative and associative properties • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

**True or false?**

$$8 + 5 = 5 + 8$$

**A.** True

**B.** False

**True or false?**

$$(2 \times 3) \times 4 = 2 \times (3 \times 4)$$

**A.** True

**B.** False

Which equation shows the commutative property of multiplication?

**A.**  $6 \times 7 = 42$

**B.**  $6 \times 7 = 7 \times 6$

**C.**  $6 \times (7 + 1) = 6 \times 7 + 6$

**D.**  $6 + 0 = 6$

**True or false?**

$$9 - 4 = 4 - 9$$

**A.** True

**B.** False

Fill in the blank.

$$3 \times 8 = 8 \times \underline{\quad}$$

Blank: \_\_\_\_\_



SET 48 • 6.EE.A.4

# BACKWARD 2

Evaluate an expression for a given value • Grade 6 • Expressions & Equations •  
6.EE.A.2.c

Evaluate  $4x + 2x$  when  $x = 3$ .

Value: \_\_\_\_\_

Evaluate  $3(x + 2)$  when  $x = 5$ .

Value: \_\_\_\_\_

## True or false?

When  $x = 2$ , the value of  $3x + 2$  is 8.

**A.** True

**B.** False

Evaluate  $2(x + 4)$  when  $x = 1$ .

- A.** 6
- B.** 9
- C.** 10
- D.** 12

Evaluate  $5a + 15$  when  $a = 2$ .

Value: \_\_\_\_\_

SET 48 • 6.EE.A.4

# BACKWARD 3

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

**True or false?**

$$6 + 6 + 6 + 6 = 4 \times 6$$

**A.** True

**B.** False



Which expression means 3 groups of 8?

**A.**  $3 + 8$

**B.**  $38$

**C.**  $8 - 3$

**D.**  $3 \times 8$

Write  $5 + 5 + 5$  as a multiplication expression.

Expression: \_\_\_\_\_

**True or false?**

$$7 \times 2 = 7 + 7$$

**A.** True

**B.** False

How many groups of 4 are in  $4 + 4 + 4 + 4 + 4$ ?

Groups: \_\_\_\_\_

SET 48 • 6.EE.A.4

# FORWARD 1

Rewrite expressions to show how quantities are related • Grade 7 •  
Expressions & Equations • 7.EE.A.2

A price  $c$  plus an 8% tax can be written as  $c + 0.08c$ .

Which expression is equivalent?

- A.**  $0.08c$
- B.**  $1.08c$
- C.**  $1.8c$
- D.**  $8.08c$

**True or false?**

$$a - 0.2a = 0.8a$$

**A.** True

**B.** False

Write an expression with one term that is equivalent to  $n - 0.25n$ .

Expression: \_\_\_\_\_



Which expression shows the price  $p$  after a 30% discount?

- A.**  $0.3p$
- B.**  $0.7p$
- C.**  $1.3p$
- D.**  $p - 30$

The perimeter of a square is  $s + s + s + s$ .

Write an equivalent expression with one term.

Expression: \_\_\_\_\_

SET 48 • 6.EE.A.4

# FORWARD 2

One, none, or infinitely many solutions • Grade 8 • Expressions & Equations •  
8.EE.C.7.a

## True or false?

The equation  $2(x + 3) = 2x + 6$  has infinitely many solutions.

**A.** True

**B.** False

How many solutions does the equation have?

$$5x - 3 = 5x + 1$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

How many solutions does the equation have?

$$3(2x + 4) = 6x + 12$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

Write a number in the box so the equation has no solution.

$$4x + 7 = 4x + \square$$

$$\square = \underline{\hspace{10cm}}$$

Which equation has exactly one solution?

**A.**  $3x + 1 = 2x + 1$

**B.**  $3x + 1 = 3x + 2$

**C.**  $3x + 1 = 3x + 1$

**D.**  $3(x + 1) = 3x + 3$



# Use substitution to decide whether a number makes an equation or inequality true

|                   |                                                             |                   |
|-------------------|-------------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 6 main question                                       | <b>6.EE.B.5</b>   |
| <b>BACKWARD 1</b> | Evaluate an expression                                      | <b>6.EE.A.2.c</b> |
| <b>BACKWARD 2</b> | Comparing numbers                                           | <b>2.NBT.A.4</b>  |
| <b>BACKWARD 3</b> | Meaning of the equal sign                                   | <b>1.OA.D.7</b>   |
| <b>FORWARD 1</b>  | Solve equations of the form $px + q = r$ and $p(x + q) = r$ | <b>7.EE.B.4.a</b> |
| <b>FORWARD 2</b>  | Solutions of a system of two linear equations               | <b>8.EE.C.8.a</b> |

SET 49 • 6.EE.B.5

# MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.5

Which value from the set  $\{2, 4, 6, 8\}$  makes  $3x + 1 = 19$  true?

**A.** 2

**B.** 4

**C.** 6

**D.** 8

## True or false?

$x = 5$  is a solution of  $2x - 3 > 8$ .

**A.** True

**B.** False

Which numbers from the set  $\{1, 3, 5, 7\}$  make  $x + 4 < 10$  true?

Numbers: \_\_\_\_\_

Is  $y = 2.5$  a solution of  $4y = 10$ ?

**A.** Yes

**B.** No

Which value from the set  $\{12, 15, 18\}$  makes  $n \div 3 = 6$  true?

$n =$  \_\_\_\_\_

SET 49 • 6.EE.B.5

# BACKWARD 1

Evaluate an expression • Grade 6 • Expressions & Equations • 6.EE.A.2.c



Evaluate  $3x + 1$  when  $x = 6$ .

Value: \_\_\_\_\_

Evaluate  $2x - 3$  when  $x = 5$ .

Value: \_\_\_\_\_

## True or false?

When  $y = 4$ , the value of  $5y$  is 20.

**A.** True

**B.** False

Evaluate  $x + 4$  when  $x = 7$ .

- A.** 3
- B.** 11
- C.** 28
- D.** 47

Evaluate  $n \div 3$  when  $n = 15$ .

Value: \_\_\_\_\_

SET 49 • 6.EE.B.5

# BACKWARD 2

Comparing numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Write  $>$ ,  $<$ , or  $=$  to compare.

19      21

Symbol: \_\_\_\_\_

# True or false?

$$7 > 8$$

**A.** True

**B.** False



Which statement is true?

**A.**  $9 > 6$

**B.**  $10 < 5$

**C.**  $11 < 10$

**D.**  $6 > 9$

**True or false?**

$$24 = 24$$

**A.** True

**B.** False

Write  $>$ ,  $<$ , or  $=$  to compare.

105      150

Symbol: \_\_\_\_\_

SET 49 • 6.EE.B.5

# BACKWARD 3

Meaning of the equal sign • Grade 1 • Operations & Algebraic Thinking •  
1.OA.D.7

**True or false?**

$$4 + 5 = 9 + 0$$

**A.** True

**B.** False

**True or false?**

$$12 = 15 - 3$$

**A.** True

**B.** False

**True or false?**

$$3 + 6 = 5 + 3$$

**A.** True

**B.** False

Which equation is true?

**A.**  $4 + 3 = 5 + 3$

**B.**  $4 + 3 = 5 + 2$

**C.**  $7 = 4 - 3$

**D.**  $4 + 3 = 8$



**True or false?**

$$10 - 2 = 7$$

**A.** True

**B.** False

SET 49 • 6.EE.B.5

# FORWARD 1

Solve equations of the form  $px + q = r$  and  $p(x + q) = r$  • Grade 7 •  
Expressions & Equations • 7.EE.B.4.a

Solve for x.

$$3x + 7 = 22$$

x = \_\_\_\_\_

Solve for x.

$$4(x - 2) = 20$$

x = \_\_\_\_\_

Solve for x.

$$0.5x + 6 = 10$$

**A.** 2

**B.** 8

**C.** 20

**D.** 32

Solve for x.

$$\frac{2}{3}x - 4 = 10$$

x = \_\_\_\_\_

A movie ticket costs \$12. Popcorn costs  $p$  dollars. Two tickets and one popcorn cost \$31.

What is  $p$ ?

$p =$  \_\_\_\_\_

SET 49 • 6.EE.B.5

# FORWARD 2

Solutions of a system of two linear equations • Grade 8 • Expressions & Equations • 8.EE.C.8.a



## True or false?

$(2, 3)$  is a solution of both  $y = x + 1$  and  $y = 2x - 1$ .

**A.** True

**B.** False

Which point is a solution of both  $y = 2x$  and  $y = x + 3$ ?

**A.** (0, 3)

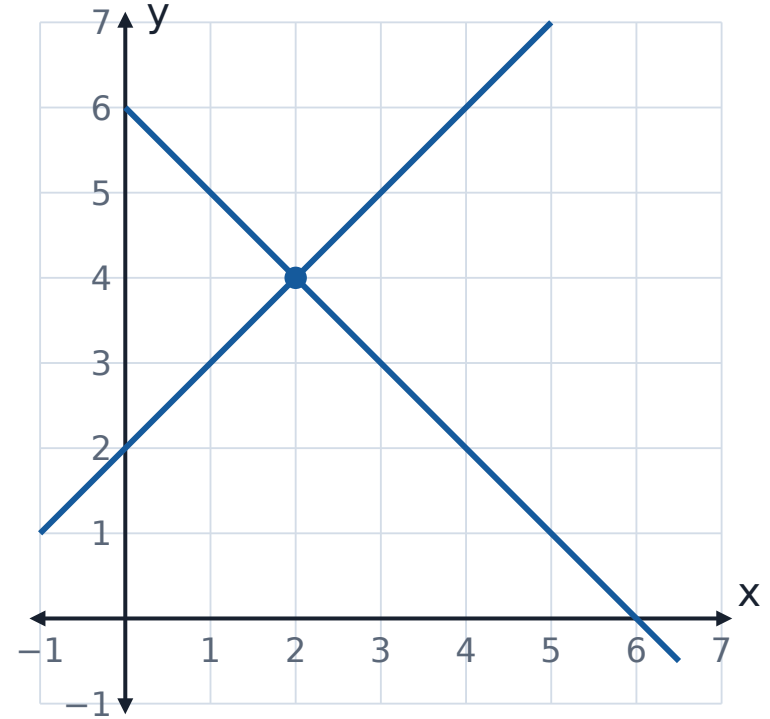
**B.** (2, 5)

**C.** (1, 2)

**D.** (3, 6)

What is the solution of the system shown by the graph?

Solution: \_\_\_\_\_



## True or false?

(1, 4) is a solution of both  $y = 3x + 1$  and  $y = -x + 5$ .

**A.** True

**B.** False

Is  $(0, 2)$  a solution of both  $y = 2x + 2$  and  $y = x - 2$ ?

**A.** Yes

**B.** No

# Use variables to write expressions for real-world problems

|                   |                                             |                   |
|-------------------|---------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 6 main question                       | <b>6.EE.B.6</b>   |
| <b>BACKWARD 1</b> | Word problems with an unknown number        | <b>3.OA.A.3</b>   |
| <b>BACKWARD 2</b> | Multiplicative comparison word problems     | <b>4.OA.A.2</b>   |
| <b>BACKWARD 3</b> | Number patterns that follow a rule          | <b>4.OA.C.5</b>   |
| <b>FORWARD 1</b>  | Write and solve equations for word problems | <b>7.EE.B.4.a</b> |
| <b>FORWARD 2</b>  | Model a linear relationship with a function | <b>8.F.B.4</b>    |

SET 50 • 6.EE.B.6

# MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.6

Tickets cost \$8 each.

Write an expression for the cost of  $t$  tickets.

Expression: \_\_\_\_\_



Kim is  $k$  years old. Sam is 3 years older than Kim.

Write an expression for Sam's age.

Expression: \_\_\_\_\_

A pizza is cut into  $s$  slices. 4 friends share the slices equally.

Which expression shows how many slices each friend gets?

**A.**  $4s$

**B.**  $s - 4$

**C.**  $\frac{s}{4}$

**D.**  $s + 4$

The length of a rectangle is 5 inches more than its width  $w$ .

Write an expression for the length.

Expression: \_\_\_\_\_

A gym charges \$20 to join plus \$15 per month.

Write an expression for the total cost after  $m$  months.

Expression: \_\_\_\_\_

SET 50 • 6.EE.B.6

# BACKWARD 1

Word problems with an unknown number • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

There are 6 bags with the same number of apples.  
There are 42 apples in all.

How many apples are in each bag?

Apples: \_\_\_\_\_

There are 5 boxes with  $\square$  pencils in each box. There are 35 pencils in all.

Which equation matches?

**A.**  $35 \times 5 = \square$

**B.**  $\square - 5 = 35$

**C.**  $5 + \square = 35$

**D.**  $5 \times \square = 35$

## True or false?

If  $4 \times \square = 28$ , then  $\square = 7$ .

**A.** True

**B.** False



There are 8 rows with  $\square$  chairs in each row. There are 56 chairs in all.

What number is  $\square$ ?

$\square =$  \_\_\_\_\_

Jan split 45 stickers equally among 9 friends.  
How many stickers did each friend get?

Stickers: \_\_\_\_\_

SET 50 • 6.EE.B.6

# BACKWARD 2

Multiplicative comparison word problems • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

A red ribbon is 7 inches long. A blue ribbon is 3 times as long.

How long is the blue ribbon?

Length: \_\_\_\_\_

Ben has 6 books. Ali has 4 times as many books as Ben.  
Which equation finds  $b$ , the number of books Ali has?

**A.**  $b = 4 \times 6$

**B.**  $b = 4 + 6$

**C.**  $b = 6 \div 4$

**D.**  $b = 6 - 4$

## True or false?

A cat weighs 5 pounds. A dog weighs 8 times as much. The dog weighs 40 pounds.

**A.** True

**B.** False

A bush is 4 feet tall. A tree is 5 times as tall.

How tall is the tree?

Height: \_\_\_\_\_

Sara ran 24 miles. That is 3 times as far as Tom ran.

How far did Tom run?

Miles: \_\_\_\_\_



SET 50 • 6.EE.B.6

# BACKWARD 3

Number patterns that follow a rule • Grade 4 • Operations & Algebraic Thinking  
• 4.OA.C.5

The rule is "add 4." The pattern starts at 3.  
Write the first five numbers.

Numbers: \_\_\_\_\_

The rule is "multiply by 2." The pattern starts at 1.  
What is the 5th number?

- A.** 8
- B.** 10
- C.** 16
- D.** 32

## True or false?

The rule "add 5," starting at 5, gives 5, 10, 15, 20.

**A.** True

**B.** False

What is the rule for this pattern?

7, 11, 15, 19, ...

Rule: \_\_\_\_\_

The rule is "add 6." The pattern starts at 2.

What is the 4th number?

Number: \_\_\_\_\_

SET 50 • 6.EE.B.6

# FORWARD 1

Write and solve equations for word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Sam has \$60. He spends \$8 each week.

Write and solve an equation to find when he will have \$20 left.

Weeks: \_\_\_\_\_



A rectangle has a perimeter of 50 cm and a length of 15 cm.

Which equation can be used to find its width  $w$ ?

**A.**  $2(15 + w) = 50$

**B.**  $2(15w) = 50$

**C.**  $15 + 2w = 50$

**D.**  $15 + w = 50$

A taxi charges \$3 plus \$2 per mile. The fare was \$17.

How many miles was the ride?

Miles: \_\_\_\_\_

You buy 4 notebooks and a \$6 backpack. You spend \$18 in all.

How much does each notebook cost?

Cost: \_\_\_\_\_

## True or false?

"Five less than twice a number is 17" can be written as  $2n - 5 = 17$ .

**A.** True

**B.** False

SET 50 • 6.EE.B.6

# FORWARD 2

Model a linear relationship with a function • Grade 8 • Functions • 8.F.B.4

A plant is 4 cm tall. It grows 1.5 cm each week.  
Write a function for its height  $h$  after  $w$  weeks.

$h =$  \_\_\_\_\_

Which situation matches  $y = 2x + 10$ , where  $y$  is dollars in savings after  $x$  weeks?

- A.** Save \$12 each week.
- B.** Start with \$10 and spend \$2 each week.
- C.** Start with \$10 and save \$2 each week.
- D.** Start with \$2 and save \$10 each week.

The table shows savings over time.

Write a function for the savings  $y$  after  $x$  weeks.

| Weeks ( $x$ )   | 0  | 1  | 2  |
|-----------------|----|----|----|
| Savings ( $y$ ) | 30 | 45 | 60 |

$y =$  \_\_\_\_\_



A rental truck costs \$50 plus \$0.25 per mile.

Write a function for the cost  $C$  of driving  $m$  miles.

$C =$  \_\_\_\_\_

## True or false?

A line through  $(0, 5)$  and  $(2, 9)$  has an initial value of 5 and a rate of change of 2.

**A.** True

**B.** False

# Solve one-step equations of the form

$$x + p = q$$

|                   |                                                    |                   |
|-------------------|----------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 6 main question                              | <b>6.EE.B.7</b>   |
| <b>BACKWARD 1</b> | Subtraction as an unknown-addend problem           | <b>1.OA.B.4</b>   |
| <b>BACKWARD 2</b> | Adding and subtracting decimals                    | <b>5.NBT.B.7</b>  |
| <b>BACKWARD 3</b> | Adding and subtracting fractions and mixed numbers | <b>5.NF.A.1</b>   |
| <b>FORWARD 1</b>  | Solve two-step equations                           | <b>7.EE.B.4.a</b> |
| <b>FORWARD 2</b>  | Solve equations with variables on both sides       | <b>8.EE.C.7.b</b> |

SET 51 • 6.EE.B.7

# MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + 7.5 = 12$$

x = \_\_\_\_\_

Solve for n.

$$n - \frac{3}{4} = 5\frac{1}{2}$$

**A.**  $4\frac{3}{4}$

**B.**  $5\frac{1}{4}$

**C.**  $6\frac{1}{4}$

**D.**  $6\frac{3}{4}$

Solve for y.

$$y + 2\frac{1}{3} = 6$$

y = \_\_\_\_\_

Solve for m.

$$18.25 + m = 30$$

m = \_\_\_\_\_



Solve for k.

$$k - 4.8 = 9.2$$

$$k = \underline{\hspace{10cm}}$$

SET 51 • 6.EE.B.7

# BACKWARD 1

Subtraction as an unknown-addend problem • Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4

## True or false?

$12 - 7 = \square$  can be solved by thinking  
 $7 + \square = 12$ .

**A.** True

**B.** False

What number goes in the box?

$$7 + \square = 15$$

$$\square = \underline{\hspace{10em}}$$

To find  $16 - 9$ , think  $9 + \square = 16$ .

What number goes in the box?

- A.** 6
- B.** 7
- C.** 8
- D.** 25

What number goes in the box?

$$\square + 8 = 13$$

$$\square = \underline{\hspace{10em}}$$

## True or false?

To find  $11 - 5$ , you can think  $5 + 6 = 11$ .

**A.** True

**B.** False

SET 51 • 6.EE.B.7

# BACKWARD 2

Adding and subtracting decimals • Grade 5 • Number & Operations in Base Ten  
• 5.NBT.B.7



Subtract.

$$12 - 7.5$$

Difference: \_\_\_\_\_

Subtract.

$$30 - 18.25$$

Difference: \_\_\_\_\_

**True or false?**

$$9.2 + 4.8 = 14$$

**A.** True

**B.** False

$$6.25 + 2.8 = ?$$

**A.** 6.53

**B.** 8.33

**C.** 9.05

**D.** 9.5

Add.

$$3.7 + 5.45$$

Sum: \_\_\_\_\_

SET 51 • 6.EE.B.7

# BACKWARD 3

Adding and subtracting fractions and mixed numbers • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$$6 - 2\frac{1}{3}$$

Difference: \_\_\_\_\_

Add.

$$5\frac{1}{2} + \frac{3}{4}$$

Sum: \_\_\_\_\_



**True or false?**

$$\frac{2}{3} + \frac{1}{6} = \frac{5}{6}$$

**A.** True

**B.** False

$$\frac{7}{8} - \frac{1}{4} = ?$$

**A.**  $\frac{1}{2}$

**B.**  $\frac{5}{8}$

**C.**  $\frac{3}{4}$

**D.**  $\frac{6}{4}$

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q4/5

Add.

$$1\frac{2}{5} + \frac{1}{2}$$

Sum: \_\_\_\_\_

SET 51 • 6.EE.B.7

# FORWARD 1

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$$3x + 8 = 29$$

x = \_\_\_\_\_

Solve for x.

$$-2x + 5 = 17$$

x = \_\_\_\_\_

Solve for x.

$$\frac{x}{4} - 3 = 2$$

**A.**     $-4$

**B.**     $\frac{5}{4}$

**C.**     $5$

**D.**     $20$

Solve for x.

$$0.6x + 1.5 = 4.5$$

x = \_\_\_\_\_



Solve for x.

$$5(x + 2) = 45$$

x = \_\_\_\_\_

SET 51 • 6.EE.B.7

# FORWARD 2

Solve equations with variables on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$7x - 4 = 3x + 12$$

x = \_\_\_\_\_

Solve for x.

$$-5x + 9 = x - 3$$

x = \_\_\_\_\_

Solve for x.

$$2.5x + 4 = 1.5x + 9$$

**A.** -5

**B.** 5

**C.** 6.5

**D.** 13

Solve for x.

$$\frac{1}{2}x + 6 = 2x - 3$$

x = \_\_\_\_\_

Solve for x.

$$3(2x - 5) = 4x + 1$$

x = \_\_\_\_\_

# Solve one-step equations of the form

$$px = q$$

|                   |                                            |                   |
|-------------------|--------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 6 main question                      | <b>6.EE.B.7</b>   |
| <b>BACKWARD 1</b> | Division as an unknown-factor problem      | <b>3.OA.B.6</b>   |
| <b>BACKWARD 2</b> | Dividing by a fraction                     | <b>6.NS.A.1</b>   |
| <b>BACKWARD 3</b> | Dividing decimals                          | <b>5.NBT.B.7</b>  |
| <b>FORWARD 1</b>  | Solve equations of the form $p(x + q) = r$ | <b>7.EE.B.4.a</b> |
| <b>FORWARD 2</b>  | Solve multi-step linear equations          | <b>8.EE.C.7.b</b> |



SET 52 • 6.EE.B.7

# MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$4x = 30$$

x = \_\_\_\_\_

Solve for y.

$$\frac{2}{3}y = 12$$

y = \_\_\_\_\_

Solve for m.

$$0.6m = 4.2$$

$$m = \underline{\hspace{10em}}$$

Solve for n.

$$5n = 3.5$$

**A.**  $-1.5$

**B.**  $0.7$

**C.**  $1.43$

**D.**  $17.5$

Solve for k.

$$\frac{3}{4}k = \frac{9}{10}$$

k = \_\_\_\_\_

SET 52 • 6.EE.B.7

# BACKWARD 1

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find  $48 \div 6$  by thinking  $6 \times \square = 48$ .

$\square =$  \_\_\_\_\_



## True or false?

$63 \div 7 = 9$  because  $7 \times 9 = 63$ .

**A.** True

**B.** False

Which fact helps you find  $45 \div 5$ ?

**A.**  $9 - 5 = 4$

**B.**  $45 \times 5 = 225$

**C.**  $5 \times 9 = 45$

**D.**  $5 + 40 = 45$

What number goes in the box?

$$\square \times 6 = 42$$

$$\square = \underline{\hspace{10em}}$$

## True or false?

$24 \div 4 = 5$  because  $4 \times 5 = 24$ .

**A.** True

**B.** False

SET 52 • 6.EE.B.7

# BACKWARD 2

Dividing by a fraction • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$12 \div \frac{2}{3}$$

Quotient: \_\_\_\_\_

Divide.

$$\frac{9}{10} \div \frac{3}{4}$$

Quotient: \_\_\_\_\_

**True or false?**

$$5 \div \frac{5}{8} = 8$$

**A.** True

**B.** False



$$\frac{9}{10} \div \frac{3}{5} = ?$$

**A.**  $\frac{27}{50}$

**B.**  $\frac{2}{3}$

**C.**  $1\frac{1}{3}$

**D.**  $1\frac{1}{2}$

Divide.

$$6 \div \frac{3}{4}$$

Quotient: \_\_\_\_\_

SET 52 • 6.EE.B.7

# BACKWARD 3

Dividing decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Divide.

$$4.2 \div 0.6$$

Quotient: \_\_\_\_\_

Divide.

$$30 \div 4$$

Quotient: \_\_\_\_\_

**True or false?**

$$3.5 \div 5 = 0.7$$

**A.** True

**B.** False

$$2.4 \div 0.3 = ?$$

**A.** 0.8

**B.** 7

**C.** 8

**D.** 80

Divide.

$$7.2 \div 9$$

Quotient: \_\_\_\_\_



SET 52 • 6.EE.B.7

# FORWARD 1

Solve equations of the form  $p(x + q) = r$  • Grade 7 • Expressions & Equations •  
7.EE.B.4.a

Solve for x.

$$3(x - 4) = 21$$

x = \_\_\_\_\_

Solve for x.

$$\frac{1}{2}(x + 6) = 9$$

x = \_\_\_\_\_

Solve for x.

$$3(x + 1.5) = 12$$

**A.** 2

**B.** 2.5

**C.** 4

**D.** 5.5

Solve for x.

$$-2(x - 3) = 14$$

x = \_\_\_\_\_

Solve for x.

$$0.5(x + 8) = 7$$

x = \_\_\_\_\_

SET 52 • 6.EE.B.7

# FORWARD 2

Solve multi-step linear equations • Grade 8 • Expressions & Equations •  
8.EE.C.7.b

Solve for x.

$$3(2x - 1) = 4x + 7$$

x = \_\_\_\_\_



Solve for x.

$$\frac{2}{3}(x + 3) = x - 1$$

x = \_\_\_\_\_

Solve for x.

$$5x - 2(x + 1) = 10$$

**A.** 2

**B.**  $\frac{8}{3}$

**C.** 4

**D.** 12

Solve for x.

$$0.4(10x - 5) = 2x + 4$$

x = \_\_\_\_\_

Solve for x.

$$6x = 2(x + 8)$$

x = \_\_\_\_\_

# Write and solve one-step equations for real-world problems

|                   |                                                        |                   |
|-------------------|--------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 6 main question                                  | <b>6.EE.B.7</b>   |
| <b>BACKWARD 1</b> | Word problems with an unknown factor                   | <b>3.OA.A.3</b>   |
| <b>BACKWARD 2</b> | Word problems with an unknown addend                   | <b>2.OA.A.1</b>   |
| <b>BACKWARD 3</b> | A letter stands for an unknown quantity                | <b>3.OA.D.8</b>   |
| <b>FORWARD 1</b>  | Write and solve two-step equations for word problems   | <b>7.EE.B.4.a</b> |
| <b>FORWARD 2</b>  | Solve real-world problems with variables on both sides | <b>8.EE.C.7.b</b> |

SET 53 • 6.EE.B.7

# MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.7

A pack of 6 pens costs \$4.50.

Write and solve an equation to find the cost  $p$  of one pen.

Equation: \_\_\_\_\_

$p =$  \_\_\_\_\_

After spending \$12.75, Sam has \$30.25 left.

Write and solve an equation to find how much money  $m$  Sam started with.

Equation: \_\_\_\_\_

$m =$  \_\_\_\_\_



A board is cut into 5 equal pieces. Each piece is 1.8 meters long.

Write and solve an equation to find the length  $L$  of the board.

Equation: \_\_\_\_\_

$L =$  \_\_\_\_\_

Maya is 1.35 m tall. She is 0.4 m taller than her brother.  
Which equation can be used to find her brother's height  $b$ ?

**A.**  $0.4b = 1.35$

**B.**  $1.35 + 0.4 = b$

**C.**  $b - 0.4 = 1.35$

**D.**  $b + 0.4 = 1.35$

A recipe uses  $\frac{3}{4}$  cup of sugar per batch. Kim used 6 cups of sugar.

Write and solve an equation to find the number of batches  $b$ .

Equation: \_\_\_\_\_

$b =$  \_\_\_\_\_

SET 53 • 6.EE.B.7

# BACKWARD 1

Word problems with an unknown factor • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

4 friends share some grapes equally. Each gets 9 grapes.

How many grapes were there?

Grapes: \_\_\_\_\_

There are 7 teams with  $\square$  players each. There are 56 players in all.

Which equation matches?

**A.**  $7 \times \square = 56$

**B.**  $56 \times 7 = \square$

**C.**  $\square - 7 = 56$

**D.**  $7 + \square = 56$

## True or false?

If  $3 \times \square = 24$ , then  $\square = 8$ .

**A.** True

**B.** False

A teacher puts 40 books into 5 equal stacks.  
How many books are in each stack?

Books: \_\_\_\_\_



Each box holds 6 eggs. There are 54 eggs.  
How many boxes are there?

Boxes: \_\_\_\_\_

SET 53 • 6.EE.B.7

# BACKWARD 2

Word problems with an unknown addend • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

Lin had some stickers. She got 15 more. Now she has 42.

How many did she have at first?

Stickers: \_\_\_\_\_

A plant grew 8 cm and is now 35 cm tall.

Which equation finds the starting height  $h$ ?

**A.**  $8h = 35$

**B.**  $h - 8 = 35$

**C.**  $h + 35 = 8$

**D.**  $h + 8 = 35$

## True or false?

Tom spent \$17 and has \$26 left, so he started with \$43.

**A.** True

**B.** False

A bus had 30 people. Some got off. Now there are 18.  
How many got off?

People: \_\_\_\_\_

Ada read 24 pages on Monday and some more on Tuesday. She read 61 pages in all.

How many pages did she read on Tuesday?

Pages: \_\_\_\_\_

SET 53 • 6.EE.B.7

# BACKWARD 3

A letter stands for an unknown quantity • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8



## True or false?

In  $5 \times n = 40$ , the letter  $n$  stands for 8.

**A.** True

**B.** False

In the equation  $p + 9 = 20$ , what number is  $p$ ?

$p =$  \_\_\_\_\_

Which equation matches "a number plus 6 is 15"?

**A.**  $n + 15 = 6$

**B.**  $n - 6 = 15$

**C.**  $6n = 15$

**D.**  $n + 6 = 15$

In the equation  $3 \times k = 27$ , what number is  $k$ ?

$k =$  \_\_\_\_\_

## True or false?

In  $m - 4 = 10$ , the letter  $m$  stands for 6.

**A.** True

**B.** False

SET 53 • 6.EE.B.7

# FORWARD 1

Write and solve two-step equations for word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A gym charges a \$25 fee plus \$15 per month. Jen paid \$130.

Write and solve an equation to find the number of months  $m$ .

$m =$  \_\_\_\_\_

A plumber charges \$45 plus \$60 per hour. A bill is \$225.  
How many hours did the plumber work?

Hours: \_\_\_\_\_



A class buys 12 tickets and pays a \$15 group fee. The total is \$159.

Which equation finds the cost  $t$  of one ticket?

**A.**  $12t - 15 = 159$

**B.**  $12t + 15 = 159$

**C.**  $12(t + 15) = 159$

**D.**  $15t + 12 = 159$

Mo saves \$18 per week and already has \$54. He wants \$180.

How many weeks  $w$  will it take?

$W =$  \_\_\_\_\_

## True or false?

A \$6 pizza plus 3 toppings costs \$10.50. Each topping costs \$1.50.

- A. True
- B. False

SET 53 • 6.EE.B.7

# FORWARD 2

Solve real-world problems with variables on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Store A charges \$20 plus \$3 per shirt. Store B charges \$5 per shirt.

Write and solve an equation to find when the costs are equal.

Shirts: \_\_\_\_\_

Ana has \$120 and spends \$8 per week. Ben has \$40 and saves \$12 per week.

After how many weeks will they have the same amount?

Weeks: \_\_\_\_\_

Taxi A charges \$4 plus \$2.50 per mile. Taxi B charges \$1 plus \$3 per mile.

For how many miles do the rides cost the same?

**A.** 3

**B.** 5

**C.** 6

**D.** 10

A candle 30 cm tall burns 1.5 cm per hour. Another candle 20 cm tall burns 0.5 cm per hour.

After how many hours are they the same height?

Hours: \_\_\_\_\_



Plan A: \$15 per month plus \$0.10 per text. Plan B: \$0.25 per text.

For how many texts do the plans cost the same?

Texts: \_\_\_\_\_

# Write an inequality $x > c$ or $x < c$ for a real-world condition

|                   |                                                                    |                   |
|-------------------|--------------------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 6 main question                                              | <b>6.EE.B.8</b>   |
| <b>BACKWARD 1</b> | Inequality symbols                                                 | <b>1.NBT.B.3</b>  |
| <b>BACKWARD 2</b> | Meaning of "at least," "at most," "more than," and "fewer than"    | <b>6.NS.C.7.b</b> |
| <b>BACKWARD 3</b> | Negative numbers in context                                        | <b>6.NS.C.5</b>   |
| <b>FORWARD 1</b>  | Write and solve inequalities for real-world problems               | <b>7.EE.B.4.b</b> |
| <b>FORWARD 2</b>  | Use two linear equations to decide when one quantity is greater... | <b>8.EE.C.8.c</b> |

SET 54 • 6.EE.B.8

# MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.8

Riders must be at least 48 inches tall.

Write an inequality for the height  $h$  of a rider.

Inequality: \_\_\_\_\_

A bag can hold at most 20 pounds.

Which inequality describes the weight  $w$  the bag can hold?

**A.**  $w \geq 20$

**B.**  $w < 20$

**C.**  $w \leq 20$

**D.**  $w > 20$

The temperature must stay below  $-3^{\circ}\text{C}$ .

Write an inequality for the temperature  $t$ .

Inequality: \_\_\_\_\_

More than 25 people must sign up for a trip to happen.  
Write an inequality for the number of people  $p$ .

Inequality: \_\_\_\_\_

Children 12 years old and younger get a discount.

Write an inequality for the age  $a$  of a child who gets the discount.

Inequality: \_\_\_\_\_



SET 54 • 6.EE.B.8

# BACKWARD 1

Inequality symbols • Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3

Which symbol makes this true?

58      85

**A.**    <

**B.**    >

**C.**    =

**True or false?**

$$30 > 29$$

**A.** True

**B.** False

Write  $>$ ,  $<$ , or  $=$  to compare.

64      46

Symbol: \_\_\_\_\_

Which symbol means "is greater than"?

**A.**  $<$

**B.**  $>$

**C.**  $=$

**True or false?**

$$17 < 12$$

**A.** True

**B.** False

SET 54 • 6.EE.B.8

# BACKWARD 2

Meaning of "at least," "at most," "more than," and "fewer than" • Grade 6 • The Number System • 6.NS.C.7.b

## True or false?

"At least 10" includes the number 10.

**A.** True

**B.** False



What does "at most 6" mean?

- A.** 6 or more
- B.** Less than 6
- C.** More than 6
- D.** 6 or less

## True or false?

"Fewer than 8" includes the number 8.

**A.** True

**B.** False

Which number is NOT "at least 12"?

- A.** 11
- B.** 12
- C.** 13
- D.** 20

## True or false?

9 is "more than 7."

**A.** True

**B.** False

SET 54 • 6.EE.B.8

# BACKWARD 3

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

Write an integer for 3 degrees below zero.

Integer: \_\_\_\_\_

**True or false?**

$-10^{\circ}\text{C}$  is colder than  $-3^{\circ}\text{C}$ .

**A.** True

**B.** False

Which number represents 15 feet below sea level?

- A.**  $-15$
- B.**  $-1.5$
- C.**  $0$
- D.**  $15$



Write an integer for a temperature of 7 degrees below zero.

Integer: \_\_\_\_\_

## True or false?

"Below  $-3^{\circ}\text{C}$ " includes  $-5^{\circ}\text{C}$ .

**A.** True

**B.** False

SET 54 • 6.EE.B.8

# FORWARD 1

Write and solve inequalities for real-world problems • Grade 7 • Expressions & Equations • 7.EE.B.4.b

You have \$50. Each ride costs \$6.

Write and solve an inequality to find how many rides  $r$  you can afford.

Solution: \_\_\_\_\_

A salesperson earns \$80 plus \$12 for each sale.

Write and solve an inequality for the number of sales  $s$  needed to earn more than \$200.

Solution: \_\_\_\_\_

A phone has 75% battery and uses 5% per hour. It must stay above 20%.

Which inequality gives the hours  $h$ ?

**A.**  $75 - 5h > 20$

**B.**  $75 + 5h > 20$

**C.**  $75 - 5h < 20$

**D.**  $5h - 75 > 20$

A van can carry at most 1,200 pounds. The driver weighs 180 pounds. Each box weighs 60 pounds.

Write and solve an inequality for the number of boxes  $b$ .

Solution: \_\_\_\_\_

## True or false?

The solution of  $3x + 2 \geq 17$  is  $x \geq 5$ .

**A.** True

**B.** False



## FORWARD 2

Use two linear equations to decide when one quantity is greater • Grade 8 •  
Expressions & Equations • 8.EE.C.8.c

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Plan A costs \$20 plus \$5 per month. Plan B costs \$35 plus \$2 per month.

After how many months does Plan A cost more than Plan B?

Months: \_\_\_\_\_

Tank A has 100 gallons and drains 6 gallons per minute.  
Tank B has 40 gallons and fills 4 gallons per minute.  
After how many minutes does Tank B hold more water?

Minutes: \_\_\_\_\_

Sara has \$50 and saves \$8 per week. Tom has \$90 and saves \$3 per week.

When will Sara have more money than Tom?

- A.** After 8 weeks
- B.** After 3 weeks
- C.** After 5 weeks
- D.** Never

Gym X charges \$60 plus \$4 per class. Gym Y charges \$12 per class.

For how many classes is Gym X cheaper?

Classes: \_\_\_\_\_

## True or false?

$y = 3x + 2$  and  $y = x + 10$  are equal when  $x = 4$ .  
For  $x > 4$ ,  $3x + 2$  is greater.

**A.** True

**B.** False

# Represent solutions of $x > c$ or $x < c$ on a number line

|                   |                                                                       |                   |
|-------------------|-----------------------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 6 main question                                                 | <b>6.EE.B.8</b>   |
| <b>BACKWARD 1</b> | Locating numbers on a number line                                     | <b>6.NS.C.6.c</b> |
| <b>BACKWARD 2</b> | Greater numbers are to the right                                      | <b>6.NS.C.7.a</b> |
| <b>BACKWARD 3</b> | Meaning of inequality symbols                                         | <b>1.NBT.B.3</b>  |
| <b>FORWARD 1</b>  | Solve and graph two-step inequalities                                 | <b>7.EE.B.4.b</b> |
| <b>FORWARD 2</b>  | Linear equations with one, no, or infinitely many solutions — near... | <b>8.EE.C.7.a</b> |

SET 55 • 6.EE.B.8

# MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.8



Graph the solutions of  $x > 3$  on the number line.



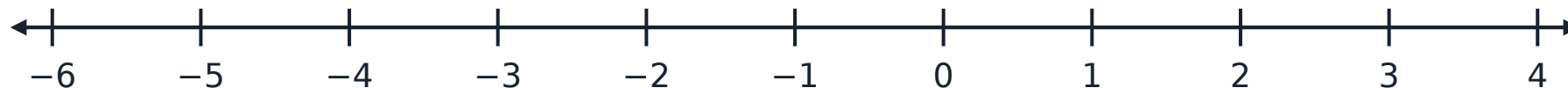
Graph the solutions of  $x \leq -1$  on the number line.



Graph the solutions of  $x < 2.5$  on the number line.



Graph the solutions of  $x \geq -4$  on the number line.



Graph the solutions of  $x > -1.5$  on the number line.



SET 55 • 6.EE.B.8

# BACKWARD 1

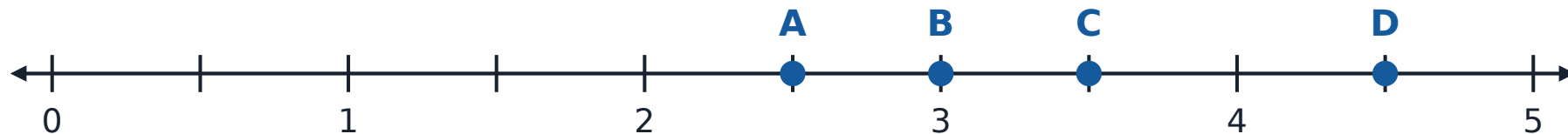
Locating numbers on a number line • Grade 6 • The Number System •  
6.NS.C.6.c

What number is at point A?



A = \_\_\_\_\_

Which point is at 3.5?



**A.** Point A

**B.** Point B

**C.** Point C

**D.** Point D



**True or false?**

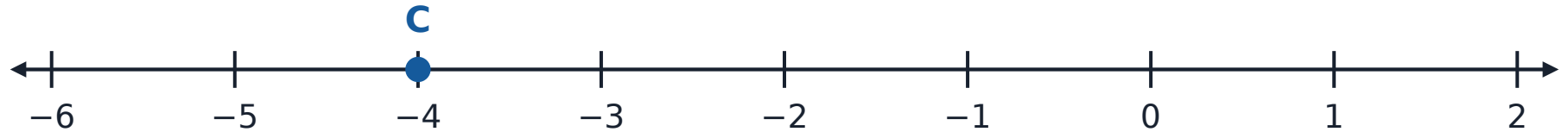
Point B is at  $-1.5$ .



**A.** True

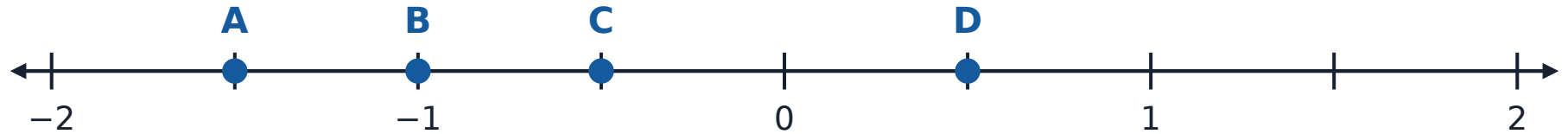
**B.** False

What number is at point C?



C = \_\_\_\_\_

Which point is at  $-\frac{1}{2}$ ?



**A.** Point A

**B.** Point B

**C.** Point C

**D.** Point D

SET 55 • 6.EE.B.8

# BACKWARD 2

Greater numbers are to the right • Grade 6 • The Number System • 6.NS.C.7.a

## True or false?

Every number to the right of 3 on a number line is greater than 3.

- A. True
- B. False

Which numbers are less than  $-1$ ?

- A.** Numbers to the right of  $-1$
- B.** Numbers to the left of  $-1$
- C.** Only  $-2$
- D.** Only  $0$

**True or false?**

$-3$  is greater than  $-1$ .

**A.** True

**B.** False

Is 2.5 to the left or the right of 2 on a number line?

Answer: \_\_\_\_\_



Which number is greater than  $-4$ ?

**A.**  $-6$

**B.**  $-5$

**C.**  $-4.5$

**D.**  $-3$

SET 55 • 6.EE.B.8

# BACKWARD 3

Meaning of inequality symbols • Grade 1 • Number & Operations in Base Ten •  
1.NBT.B.3

What does the symbol  $>$  mean?

- A.** Is equal to
- B.** Is greater than
- C.** Is less than

## True or false?

The symbol  $<$  means "is less than."

- A. True
- B. False

Which statement is true?

**A.**  $8 < 5$

**B.**  $8 = 5$

**C.**  $8 > 5$

Write the symbol for "is less than."

Symbol: \_\_\_\_\_

**True or false?**

$$12 > 21$$

**A.** True

**B.** False

SET 55 • 6.EE.B.8

# FORWARD 1

Solve and graph two-step inequalities • Grade 7 • Expressions & Equations •  
7.EE.B.4.b



Solve  $2x + 1 > 7$ . Then graph the solutions.



Solution: \_\_\_\_\_

Solve  $4x - 3 \leq 5$ . Then graph the solutions.



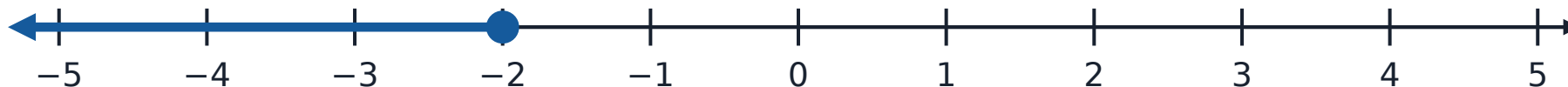
Solution: \_\_\_\_\_

Solve  $-2x + 4 < 10$ . Then graph the solutions.



Solution: \_\_\_\_\_

Which inequality does the graph show?



**A.**  $x < -2$

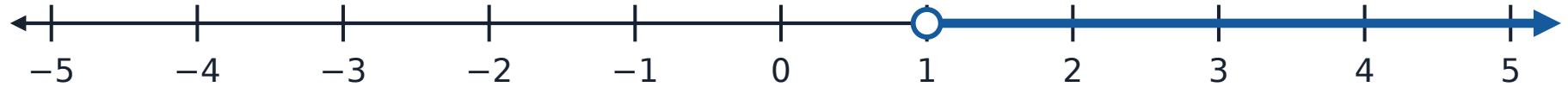
**B.**  $x > -2$

**C.**  $x \geq -2$

**D.**  $x \leq -2$

## True or false?

The graph shows the solutions of  $3x + 2 > 5$ .



**A.** True

**B.** False

## FORWARD 2

Linear equations with one, no, or infinitely many solutions • Grade 8 •  
Expressions & Equations • 8.EE.C.7.a

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

## True or false?

$x + 3 = x + 3$  has infinitely many solutions.

**A.** True

**B.** False

How many solutions does the equation have?

$$2x + 5 = 2x - 1$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions



Which equation has infinitely many solutions?

**A.**  $4x - 1 = x$

**B.**  $4x = 0$

**C.**  $4(x - 1) = 4x - 4$

**D.**  $4(x - 1) = 4x - 1$

How many solutions does the equation have?

$$3x - 7 = 3x - 7$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

How many solutions does the equation have?

$$6x - 2 = 4x + 8$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

# Identify the independent and dependent variables

**MAIN** Grade 6 main question

**6.EE.C.9**

**BACKWARD 1** Input-output rules

**4.OA.C.5**

**BACKWARD 2** Two related number patterns

**5.OA.B.3**

**FORWARD 1** Write an equation for a proportional relationship

**7.RP.A.2.c**

**FORWARD 2** Functions: each input has exactly one output

**8.F.A.1**

SET 56 • 6.EE.C.9

# MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.C.9

The equation  $c = 2.5p$  gives the cost  $c$  of  $p$  pounds of grapes.

Which is the independent variable?

- A.**  $p$  (pounds)
- B.**  $c$  (cost)
- C.** 2.5
- D.** There is none.

Money earned  $m$  depends on the number of hours worked  $h$ .

Which is the dependent variable?

Dependent variable: \_\_\_\_\_

A plant's height depends on the number of days it grows.

Which is the independent variable?

- A.** The plant's height
- B.** The number of days
- C.** Neither
- D.** Both



In  $y = 4x + 3$ , which variable is the dependent variable?

Dependent variable: \_\_\_\_\_

## True or false?

Ice cream sales  $s$  depend on the temperature  $t$ .  
So  $t$  is the independent variable.

- A. True
- B. False

SET 56 • 6.EE.C.9

# BACKWARD 1

Input-output rules • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

The rule is "multiply by 3."

What is the output when the input is 7?

Output: \_\_\_\_\_

## True or false?

The rule is "add 6." When the input is 4, the output is 10.

**A.** True

**B.** False

What is the rule?

|            |   |   |    |
|------------|---|---|----|
| <b>In</b>  | 2 | 4 | 6  |
| <b>Out</b> | 6 | 8 | 10 |

**A.** Multiply by 2

**B.** Multiply by 3

**C.** Add 2

**D.** Add 4

The rule is "subtract 2."

What is the output when the input is 15?

Output: \_\_\_\_\_

## True or false?

In an input-output table, the output depends on the input.

**A.** True

**B.** False



SET 56 • 6.EE.C.9

# BACKWARD 2

Two related number patterns • Grade 5 • Operations & Algebraic Thinking •  
5.OA.B.3

## True or false?

Pattern A: 0, 2, 4, 6

Pattern B: 0, 6, 12, 18

Each number in Pattern B is 3 times the matching number in Pattern A.

**A.** True

**B.** False

The rule is "add 5," starting at 0.

What number comes next? 0, 5, 10, \_\_\_\_

Number: \_\_\_\_\_

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 9.

When Pattern A is 12, what is the matching number in Pattern B?

Number: \_\_\_\_\_

Pattern A starts at 0 and adds 1. Pattern B starts at 0 and adds 4.

Which ordered pairs come from the patterns?

**A.** (0, 4), (1, 5), (2, 6)

**B.** (0, 0), (1, 4), (2, 8)

**C.** (1, 4), (2, 5), (3, 6)

**D.** (0, 0), (4, 1), (8, 2)

## True or false?

Pattern A: 0, 1, 2, 3

Pattern B: 0, 5, 10, 15

Each number in Pattern B is 5 more than the matching number in Pattern A.

**A.** True

**B.** False

SET 56 • 6.EE.C.9

# FORWARD 1

Write an equation for a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Tomatoes cost \$3.20 per pound.

Write an equation for the cost  $c$  of  $p$  pounds.

Equation: \_\_\_\_\_



The table shows a proportional relationship.  
Write an equation for the cost  $c$  of  $n$  ounces.

|                                |        |        |
|--------------------------------|--------|--------|
| <b>Ounces (<math>n</math>)</b> | 4      | 10     |
| <b>Cost (<math>c</math>)</b>   | \$1.20 | \$3.00 |

Equation: \_\_\_\_\_

Which equation represents a proportional relationship?

**A.**  $y = 4.5x$

**B.**  $y = 4.5x + 1$

**C.**  $y = x + 4.5$

**D.**  $y = 4.5$

A car uses 3 gallons of gas to travel 84 miles.

Write an equation for the miles  $m$  traveled on  $g$  gallons.

Equation: \_\_\_\_\_

## True or false?

$y = \frac{2}{3}x$  represents a proportional relationship.

**A.** True

**B.** False

SET 56 • 6.EE.C.9

# FORWARD 2

Functions: each input has exactly one output • Grade 8 • Functions • 8.F.A.1

## True or false?

A function assigns exactly one output to each input.

**A.** True

**B.** False

Which table does NOT represent a function?

**A.** x: 0, 1, 2 y: 0, 2, 4

**B.** x: 1, 1, 2 y: 3, 5, 7

**C.** x: 2, 3, 4 y: 5, 6, 7

**D.** x: 1, 2, 3 y: 4, 4, 4

In the function  $C = 12t$ , which variable is the input?

Input: \_\_\_\_\_



## True or false?

The set of ordered pairs  $(1, 2)$ ,  $(2, 3)$ ,  $(1, 4)$  represents a function.

- A. True
- B. False

A function gives the height of a ball after  $t$  seconds.  
What is the input?

- A.** The speed
- B.** The ball
- C.** The height
- D.** The time  $t$

# Write an equation for the relationship shown in a table or graph

|                   |                                                                 |                   |
|-------------------|-----------------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 6 main question                                           | <b>6.EE.C.9</b>   |
| <b>BACKWARD 1</b> | Finding a rule from a table                                     | <b>4.OA.C.5</b>   |
| <b>BACKWARD 2</b> | Reading points on a graph                                       | <b>5.G.A.2</b>    |
| <b>BACKWARD 3</b> | Ordered pairs from two patterns                                 | <b>5.OA.B.3</b>   |
| <b>FORWARD 1</b>  | Equations for proportional relationships from tables and graphs | <b>7.RP.A.2.c</b> |
| <b>FORWARD 2</b>  | Write a linear function from a table or graph                   | <b>8.F.B.4</b>    |

SET 57 • 6.EE.C.9

# MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.C.9

Write an equation that shows how  $y$  depends on  $x$ .

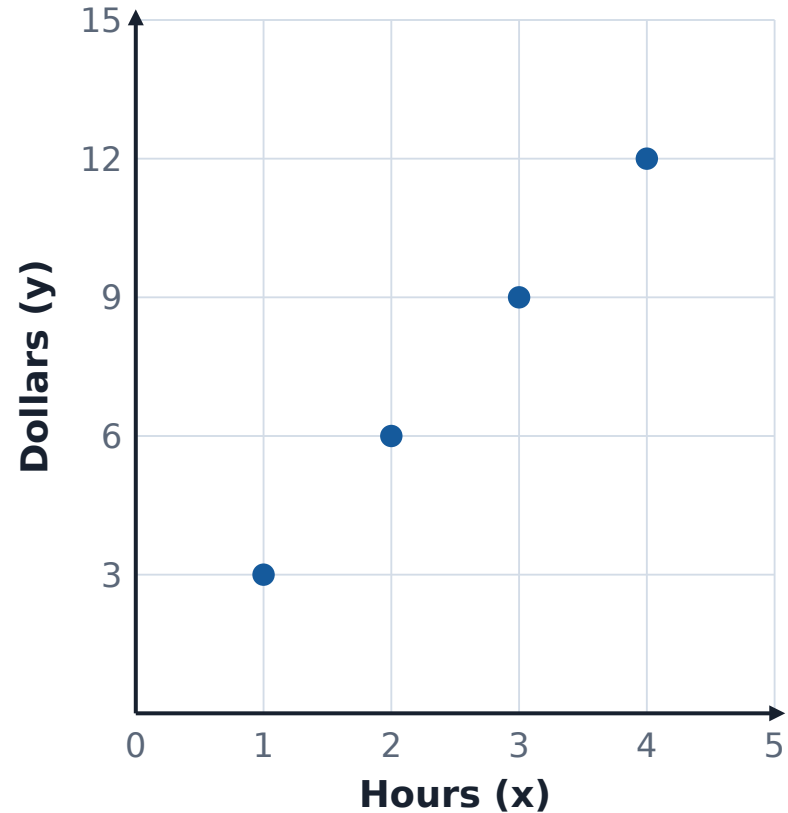
|          |   |    |    |    |
|----------|---|----|----|----|
| <b>x</b> | 1 | 2  | 3  | 4  |
| <b>y</b> | 6 | 12 | 18 | 24 |

$y =$  \_\_\_\_\_

The graph shows  $y$  dollars earned for  $x$  hours worked.

Write an equation for the relationship.

$y =$  \_\_\_\_\_



The table shows a plant's height  $h$  after  $w$  weeks.  
Write an equation for  $h$  in terms of  $w$ .

| Weeks ( $w$ )  | 0 | 1 | 2 | 3  |
|----------------|---|---|---|----|
| Height ( $h$ ) | 5 | 7 | 9 | 11 |

$h =$  \_\_\_\_\_

Which equation matches the table?

|          |   |   |   |
|----------|---|---|---|
| <b>x</b> | 1 | 2 | 3 |
| <b>y</b> | 4 | 5 | 6 |

**A.**  $y = 4x$

**B.**  $y = x + 4$

**C.**  $y = x + 3$

**D.**  $y = 3x$



The table shows the cost  $c$  of  $n$  bags of popcorn.  
Write an equation for  $c$  in terms of  $n$ .

|                              |   |    |    |
|------------------------------|---|----|----|
| <b>Bags (<math>n</math>)</b> | 2 | 4  | 6  |
| <b>Cost (<math>c</math>)</b> | 7 | 14 | 21 |

$C =$  \_\_\_\_\_

SET 57 • 6.EE.C.9

# BACKWARD 1

Finding a rule from a table • Grade 4 • Operations & Algebraic Thinking •  
4.OA.C.5

What is the rule?

|            |   |    |    |    |
|------------|---|----|----|----|
| <b>In</b>  | 1 | 2  | 3  | 4  |
| <b>Out</b> | 5 | 10 | 15 | 20 |

Rule: \_\_\_\_\_

What is the rule?

|            |   |   |    |
|------------|---|---|----|
| <b>In</b>  | 3 | 5 | 7  |
| <b>Out</b> | 7 | 9 | 11 |

Rule: \_\_\_\_\_

## True or false?

The rule for the table is "multiply by 4."

|            |   |   |    |
|------------|---|---|----|
| <b>In</b>  | 1 | 2 | 3  |
| <b>Out</b> | 4 | 8 | 12 |

**A.** True

**B.** False

What is the rule?

|            |   |   |   |
|------------|---|---|---|
| <b>In</b>  | 3 | 6 | 9 |
| <b>Out</b> | 1 | 2 | 3 |

**A.** Multiply by 3

**B.** Subtract 2

**C.** Add 3

**D.** Divide by 3

The rule is "multiply by 6." What is the output when the input is 5?

Output: \_\_\_\_\_

SET 57 • 6.EE.C.9

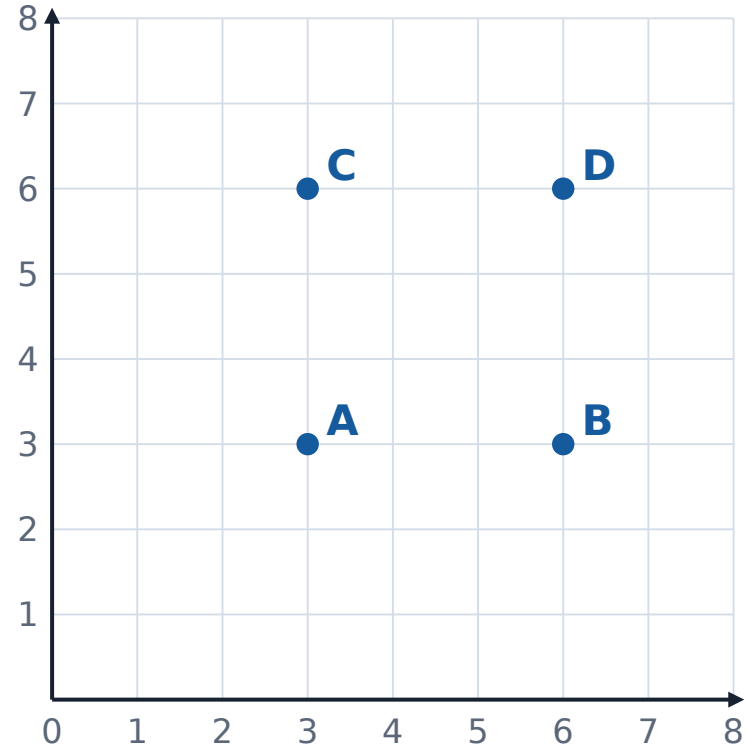
# BACKWARD 2

Reading points on a graph • Grade 5 • Geometry • 5.G.A.2



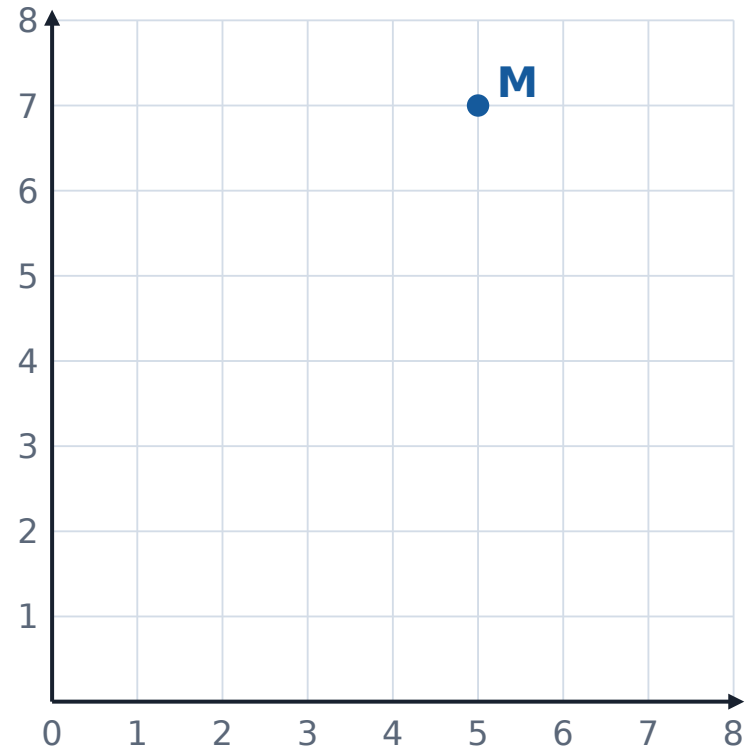
Which point is at (3, 6)?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D



What are the coordinates of point M?

M = \_\_\_\_\_



## True or false?

The point  $(2, 8)$  is 2 units right and 8 units up from the origin.

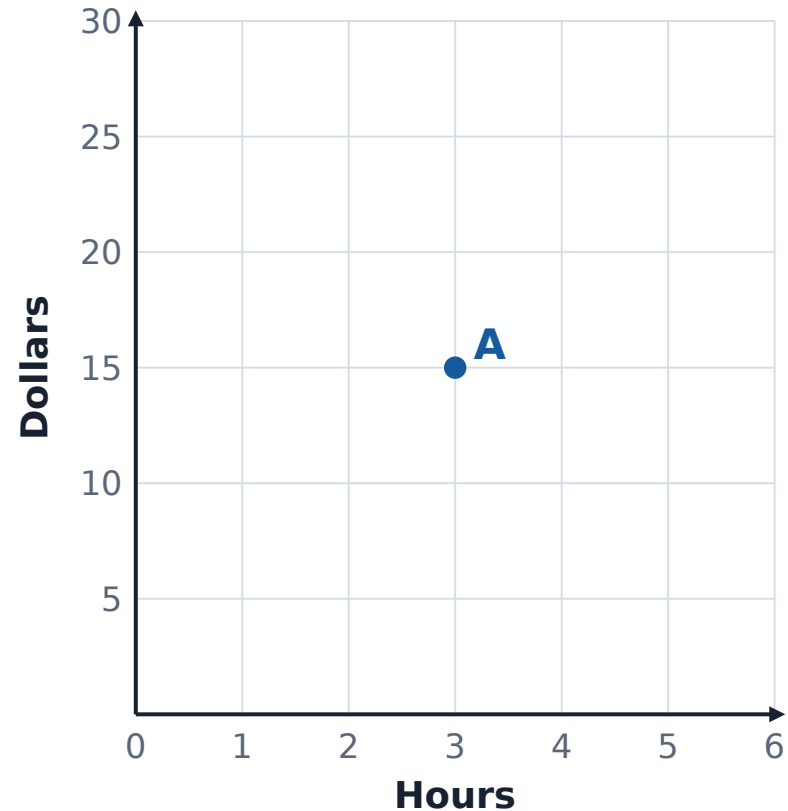
**A.** True

**B.** False

The graph shows dollars earned after working some hours.

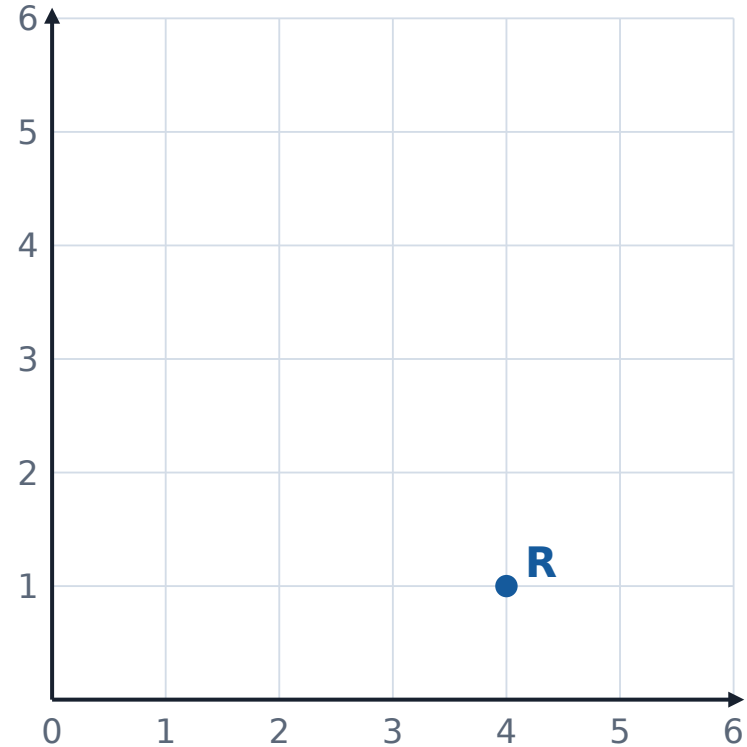
How many dollars were earned after 3 hours?

Dollars: \_\_\_\_\_



Which ordered pair names point R?

- A.** (4, 4)
- B.** (1, 4)
- C.** (1, 1)
- D.** (4, 1)



SET 57 • 6.EE.C.9

# BACKWARD 3

Ordered pairs from two patterns • Grade 5 • Operations & Algebraic Thinking •  
5.OA.B.3

Pattern A: 0, 1, 2, 3

Pattern B: 0, 5, 10, 15

Write the ordered pair formed by the 4th numbers.

Ordered pair: \_\_\_\_\_

## True or false?

The pairs (1, 3), (2, 6), (3, 9) show that each y-value is 3 times the x-value.

**A.** True

**B.** False



Pattern A adds 2. Pattern B adds 6. Both start at 0.

How does each B number compare to its A number?

- A.** 2 more
- B.** 3 times as large
- C.** 4 more
- D.** 6 times as large

Pattern A: 0, 2, 4

Pattern B: 0, 14, 28

What is the rule that changes each A number into its B number?

Rule: \_\_\_\_\_

## True or false?

The pairs  $(0, 0)$ ,  $(1, 4)$ ,  $(2, 8)$  come from Pattern A "add 4" and Pattern B "add 1."

- A. True
- B. False

SET 57 • 6.EE.C.9

# FORWARD 1

Equations for proportional relationships from tables and graphs • Grade 7 •  
Ratios & Proportional Relationships • 7.RP.A.2.c

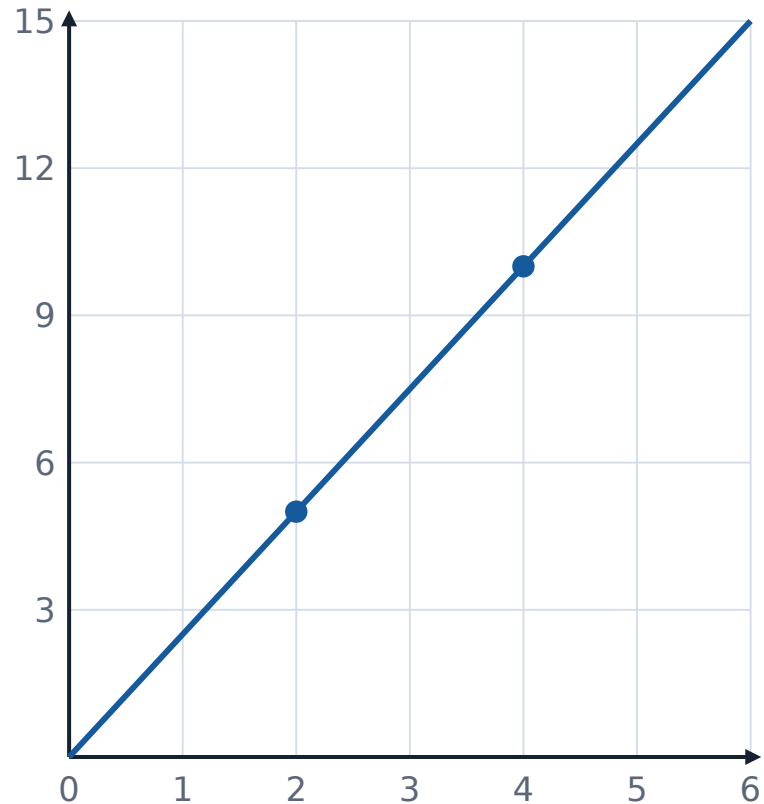
Write an equation for the proportional relationship in the table.

|          |   |      |    |
|----------|---|------|----|
| <b>x</b> | 2 | 5    | 8  |
| <b>y</b> | 7 | 17.5 | 28 |

$y =$  \_\_\_\_\_

Write an equation for the proportional relationship in the graph.

$y =$  \_\_\_\_\_



The table shows a proportional relationship.  
Which equation matches?

|          |   |   |    |
|----------|---|---|----|
| <b>x</b> | 4 | 8 | 12 |
| <b>y</b> | 3 | 6 | 9  |

**A.**  $y = 0.75x$

**B.**  $y = 1.33x$

**C.**  $y = 3x$

**D.**  $y = x - 1$

Write an equation for the cost  $c$  of  $p$  pounds.

|                                |      |       |       |
|--------------------------------|------|-------|-------|
| <b>Pounds (<math>p</math>)</b> | 3    | 6     | 9     |
| <b>Cost (<math>c</math>)</b>   | 7.50 | 15.00 | 22.50 |

$C =$  \_\_\_\_\_



## True or false?

The table shows the equation  $y = 6x$ .

|          |   |    |    |
|----------|---|----|----|
| <b>x</b> | 1 | 2  | 4  |
| <b>y</b> | 6 | 12 | 24 |

**A.** True

**B.** False

SET 57 • 6.EE.C.9

# FORWARD 2

Write a linear function from a table or graph • Grade 8 • Functions • 8.F.B.4

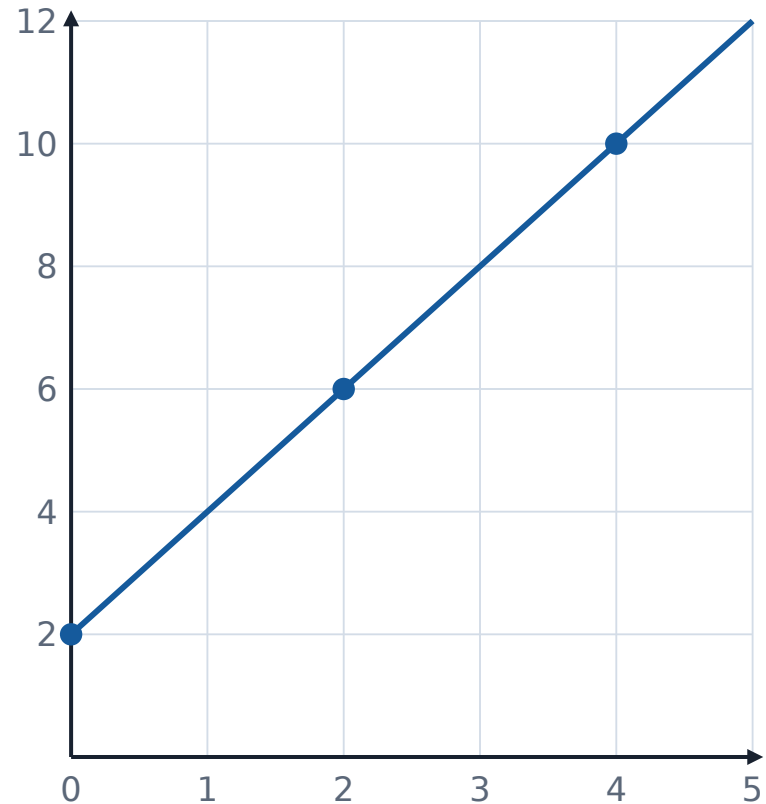
Write a function for the table.

|          |   |   |    |    |
|----------|---|---|----|----|
| <b>x</b> | 0 | 1 | 2  | 3  |
| <b>y</b> | 4 | 7 | 10 | 13 |

$y =$  \_\_\_\_\_

Write a function for the line in the graph.

$y =$  \_\_\_\_\_



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q2/5

Which function matches the table?

|          |    |   |   |
|----------|----|---|---|
| <b>x</b> | 0  | 1 | 2 |
| <b>y</b> | 10 | 8 | 6 |

**A.**  $y = -2x + 10$

**B.**  $y = 10x - 2$

**C.**  $y = -2x$

**D.**  $y = 2x + 10$

Write a function for the table.

|          |   |    |    |    |
|----------|---|----|----|----|
| <b>x</b> | 1 | 2  | 3  | 4  |
| <b>y</b> | 9 | 13 | 17 | 21 |

$y =$  \_\_\_\_\_

## True or false?

The line through  $(0, 1)$  and  $(3, 7)$  has the equation  $y = 2x + 1$ .

**A.** True

**B.** False

# Area of right triangles and other triangles

## MAIN

Grade 6 main question

**6.G.A.1**

## BACKWARD 1

Area of rectangles

**3.MD.C.7.b**

## BACKWARD 2

Half of a product

**5.NF.B.4.a**

## BACKWARD 3

Right angles and perpendicular segments

**4.G.A.1**

## FORWARD 1

Area and volume of figures made of triangles

**7.G.B.6**

## FORWARD 2

Use the Pythagorean Theorem in right triangles

**8.G.B.7**



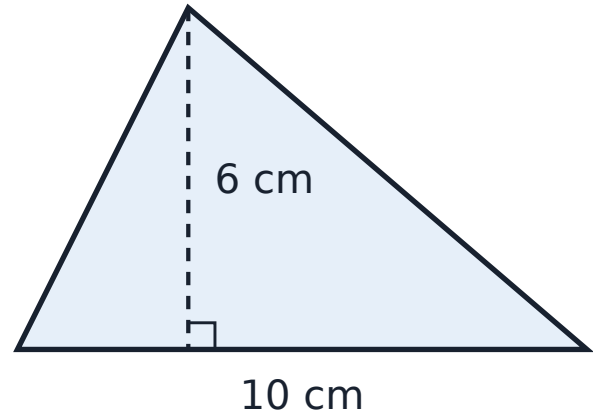
SET 58 • 6.G.A.1

# MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.1

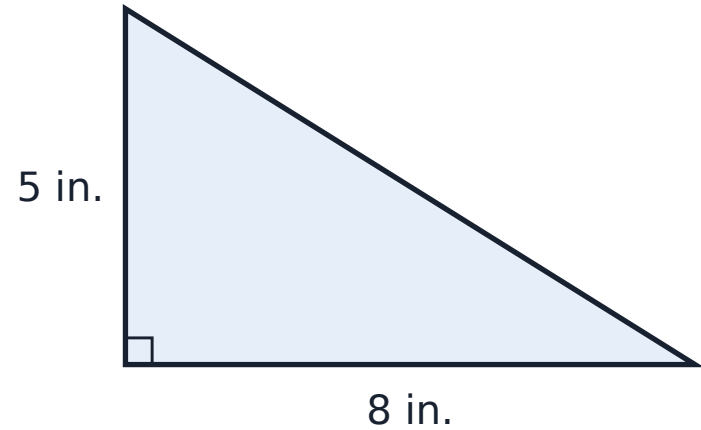
Find the area of the triangle.

Area: \_\_\_\_\_



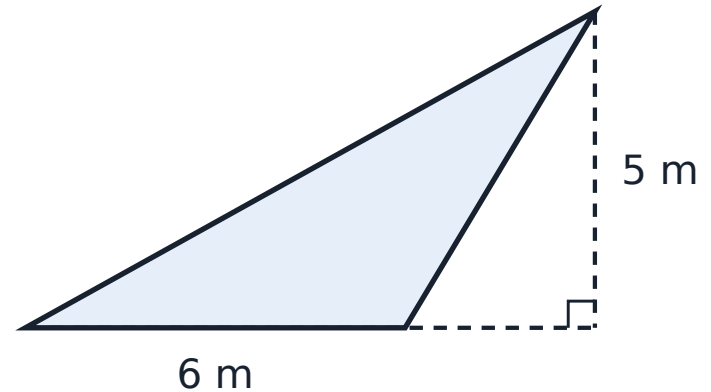
Find the area of the right triangle.

Area: \_\_\_\_\_



Find the area of the triangle.

Area: \_\_\_\_\_



A triangle has a base of 12 feet and a height of 7 feet.

What is its area?

- A.** 19 square feet
- B.** 38 square feet
- C.** 42 square feet
- D.** 84 square feet

A triangular sign has a base of 3.5 feet and a height of 2 feet.

What is its area?

Area: \_\_\_\_\_

SET 58 • 6.G.A.1

# BACKWARD 1

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

Find the area of the rectangle.

Area: \_\_\_\_\_





A rectangle is 9 feet long and 6 feet wide.

What is its area?

Area: \_\_\_\_\_

## True or false?

A rectangle that is 5 cm long and 3 cm wide has an area of 15 square centimeters.

- A. True
- B. False

A rectangle is 12 m long and 4 m wide.

What is its area?

- A.** 16 square meters
- B.** 24 square meters
- C.** 32 square meters
- D.** 48 square meters

Find the area of the rectangle.

Area: \_\_\_\_\_



SET 58 • 6.G.A.1

# BACKWARD 2

Half of a product • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{1}{2} \times 10 \times 6$$

Product: \_\_\_\_\_

Multiply.

$$\frac{1}{2} \times 14$$

Product: \_\_\_\_\_

**True or false?**

$$\frac{1}{2} \times 8 \times 5 = 20$$

**A.** True

**B.** False



$$\frac{1}{2} \times 12 \times 7 = ?$$

**A.** 19

**B.** 21

**C.** 42

**D.** 84

Multiply.

$$\frac{1}{2} \times 9 \times 4$$

Product: \_\_\_\_\_

SET 58 • 6.G.A.1

# BACKWARD 3

Right angles and perpendicular segments • Grade 4 • Geometry • 4.G.A.1

## True or false?

Perpendicular lines meet to form right angles.

**A.** True

**B.** False

Which angle measure is a right angle?

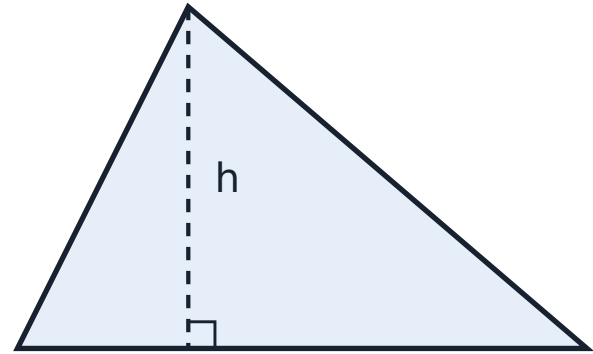
- A.**  $45^\circ$
- B.**  $90^\circ$
- C.**  $100^\circ$
- D.**  $180^\circ$

## True or false?

The dashed segment is perpendicular to the base of the triangle.

**A.** True

**B.** False



How many degrees are in a right angle?

Degrees: \_\_\_\_\_

## True or false?

A square has 4 right angles.

**A.** True

**B.** False



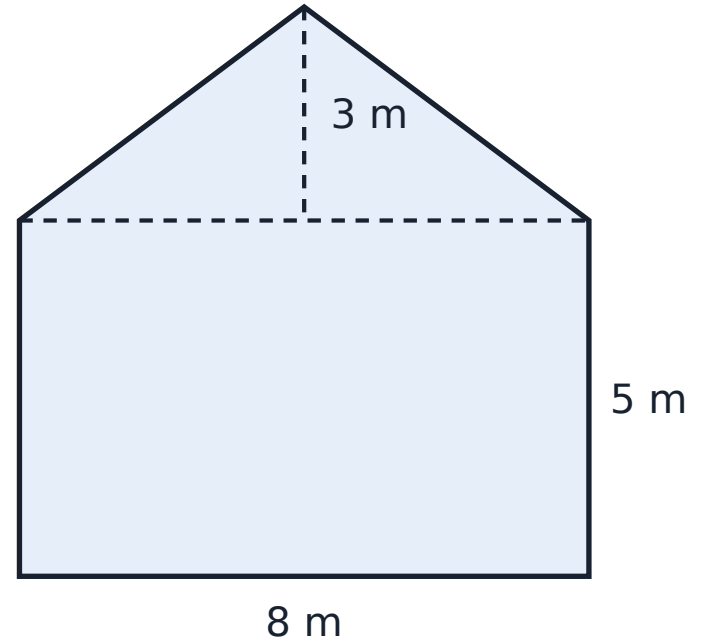
SET 58 • 6.G.A.1

# FORWARD 1

Area and volume of figures made of triangles • Grade 7 • Geometry • 7.G.B.6

Find the area of the figure.

Area: \_\_\_\_\_



A kite is made of two triangles. Each triangle has a base of 10 inches and a height of 4 inches.

What is the area of the kite?

Area: \_\_\_\_\_

A triangle with a base of 4 cm and a height of 3 cm is cut out of a rectangle that is 12 cm by 6 cm.

What is the area of the part that is left?

- A.** 60 square cm
- B.** 66 square cm
- C.** 72 square cm
- D.** 78 square cm

A triangular prism has triangle bases with a base of 6 cm and a height of 4 cm. The prism is 10 cm long.

What is its volume?

Volume: \_\_\_\_\_

## True or false?

A figure made of a 6-by-4 rectangle and a triangle with a base of 3 and a height of 4 has an area of 30 square units.

- A.** True
- B.** False

SET 58 • 6.G.A.1

# FORWARD 2

Use the Pythagorean Theorem in right triangles • Grade 8 • Geometry • 8.G.B.7

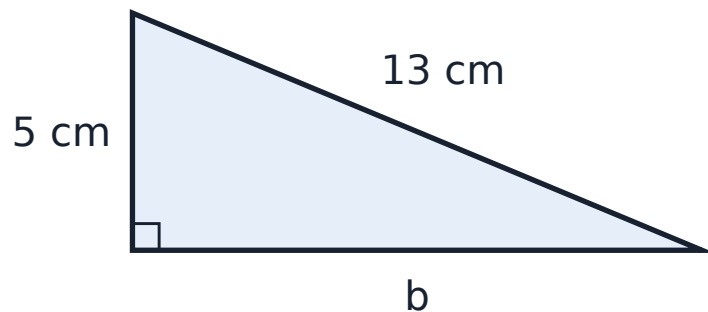
A right triangle has legs of 6 cm and 8 cm.  
What is the length of the hypotenuse?

Hypotenuse: \_\_\_\_\_



Find the length  $b$ .

$b =$  \_\_\_\_\_



A 10-foot ladder leans against a wall. The top reaches 8 feet up the wall.

How far is the bottom of the ladder from the wall?

**A.** 2 feet

**B.** 6 feet

**C.** 12.8 feet

**D.** 18 feet

An isosceles triangle has a base of 16 inches and two sides of 10 inches.

What is its area?

Area: \_\_\_\_\_

A right triangle has legs of 9 m and 12 m.  
What is the length of the hypotenuse?

Hypotenuse: \_\_\_\_\_

# Area of parallelograms and trapezoids

|                   |                                                               |                   |
|-------------------|---------------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 6 main question                                         | <b>6.G.A.1</b>    |
| <b>BACKWARD 1</b> | Area of rectangles                                            | <b>3.MD.C.7.b</b> |
| <b>BACKWARD 2</b> | Area by splitting into rectangles                             | <b>3.MD.C.7.d</b> |
| <b>BACKWARD 3</b> | Classifying quadrilaterals                                    | <b>5.G.B.4</b>    |
| <b>BACKWARD 4</b> | Height is perpendicular to the base                           | <b>4.G.A.1</b>    |
| <b>FORWARD 1</b>  | Area of figures made of quadrilaterals in real-world problems | <b>7.G.B.6</b>    |
| <b>FORWARD 2</b>  | Use the Pythagorean Theorem to find a missing height          | <b>8.G.B.7</b>    |

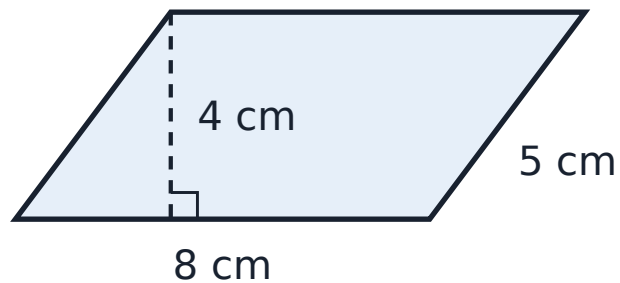
SET 59 • 6.G.A.1

# MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.1

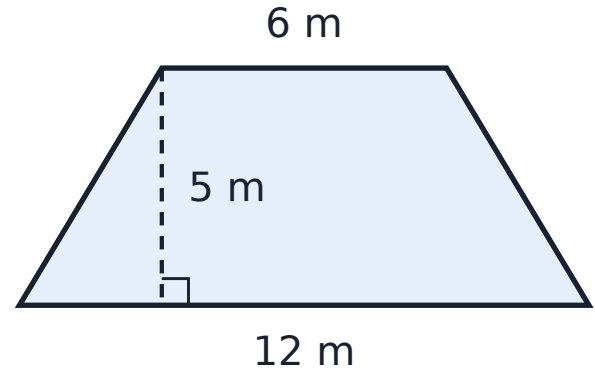
Find the area of the  
parallelogram.

Area: \_\_\_\_\_



Find the area of the trapezoid.

Area: \_\_\_\_\_





A parallelogram has a base of 9 feet and a height of 5 feet.

What is its area?

**A.** 14 square feet

**B.** 22.5 square feet

**C.** 28 square feet

**D.** 45 square feet

A trapezoid has bases of 7 inches and 11 inches. Its height is 4 inches.

What is its area?

Area: \_\_\_\_\_

A parallelogram has a base of 12.5 meters and a height of 4 meters.

What is its area?

Area: \_\_\_\_\_

SET 59 • 6.G.A.1

# BACKWARD 1

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

Find the area of the rectangle.

Area: \_\_\_\_\_



A rectangle is 11 cm long and 5 cm wide.  
What is its area?

Area: \_\_\_\_\_

## True or false?

A square with 6-meter sides has an area of 24 square meters.

**A.** True

**B.** False

A rectangle is 15 inches long and 2 inches wide.

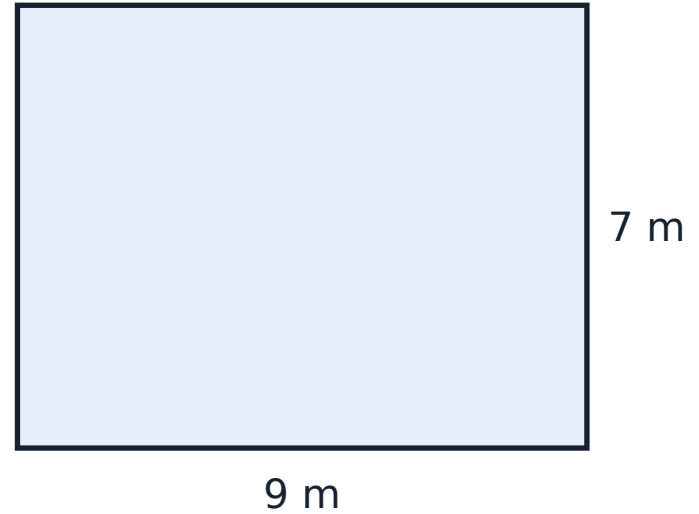
What is its area?

- A.** 17 square inches
- B.** 30 square inches
- C.** 34 square inches
- D.** 60 square inches



Find the area of the rectangle.

Area: \_\_\_\_\_



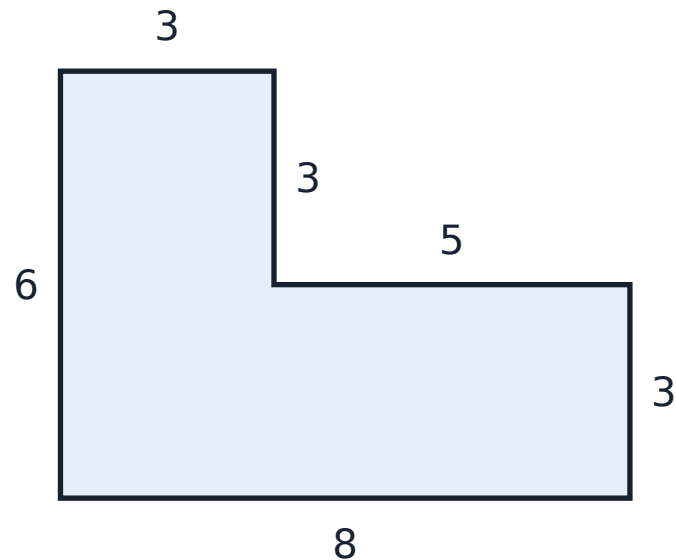
SET 59 • 6.G.A.1

# BACKWARD 2

Area by splitting into rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.d

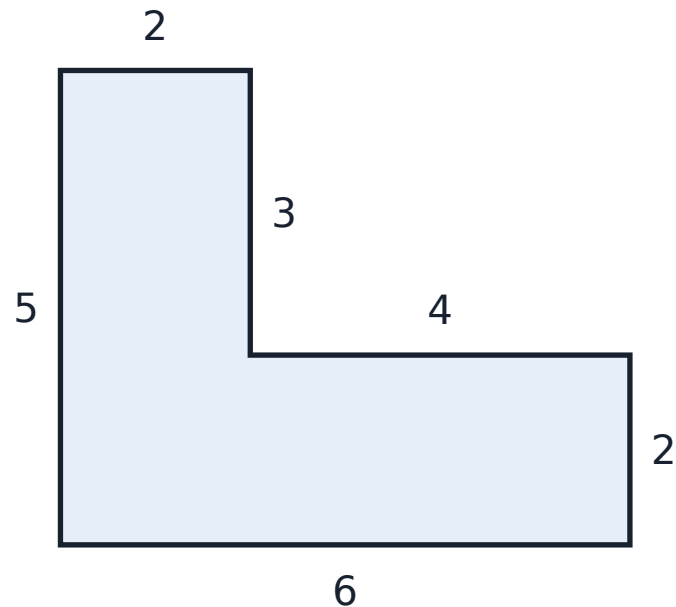
Find the area of the figure.

Area: \_\_\_\_\_



Find the area of the figure.

Area: \_\_\_\_\_



## True or false?

The area of an L-shaped figure can be found by adding the areas of two rectangles.

**A.** True

**B.** False

A figure is made of a 4-by-5 rectangle and a 2-by-3 rectangle that do not overlap.

What is its area?

**A.** 14 square units

**B.** 20 square units

**C.** 26 square units

**D.** 120 square units

A figure is made of a 10-by-2 rectangle and a 3-by-3 square that do not overlap.

What is its area?

Area: \_\_\_\_\_

SET 59 • 6.G.A.1

# BACKWARD 3

Classifying quadrilaterals • Grade 5 • Geometry • 5.G.B.4



## True or false?

Every rectangle is a parallelogram.

**A.** True

**B.** False

**True or false?**

Every parallelogram is a rectangle.

**A.** True

**B.** False

Which quadrilateral always has four right angles?

- A.** Trapezoid
- B.** Rhombus
- C.** Parallelogram
- D.** Rectangle

How many pairs of parallel sides does a parallelogram have?

**A.** 0

**B.** 1

**C.** 2

**D.** 4

## True or false?

A square is a rhombus.

**A.** True

**B.** False

SET 59 • 6.G.A.1

# BACKWARD 4

Height is perpendicular to the base • Grade 4 • Geometry • 4.G.A.1

## True or false?

The height of a parallelogram is perpendicular to its base.

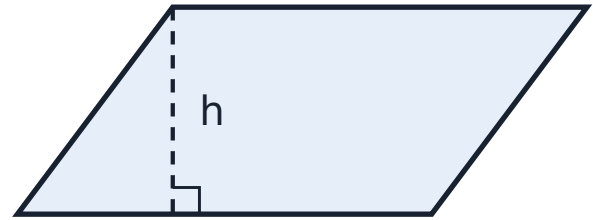
- A.** True
- B.** False

## True or false?

The dashed segment shows the height of the parallelogram.

**A.** True

**B.** False





Which statement describes perpendicular lines?

- A.** They meet at right angles.
- B.** They never meet.
- C.** They meet at a  $45^\circ$  angle.
- D.** They are always the same length.

## True or false?

The slanted side of a parallelogram is always its height.

- A.** True
- B.** False

What is the measure of the angle between a base and its height?

Angle: \_\_\_\_\_

SET 59 • 6.G.A.1

# FORWARD 1

Area of figures made of quadrilaterals in real-world problems • Grade 7 •  
Geometry • 7.G.B.6

A patio is made of a rectangle that is 10 ft by 6 ft and a trapezoid with bases of 6 ft and 4 ft and a height of 3 ft.

What is the area of the patio?

Area: \_\_\_\_\_

A wall is 12 ft by 8 ft. It has a window shaped like a parallelogram with a base of 3 ft and a height of 2 ft.

What is the area of the wall that is not window?

Area: \_\_\_\_\_

A sign is a trapezoid with bases of 1.5 m and 2.5 m and a height of 1.2 m.

What is its area?

- A.** 2.4 square meters
- B.** 3.6 square meters
- C.** 4.8 square meters
- D.** 5.2 square meters

A garden is made of two parallelograms that do not overlap. Each has a base of 7 m and a height of 4 m.

What is the total area?

Area: \_\_\_\_\_



## True or false?

A trapezoid with bases of 6 and 10 and a height of 5 has the same area as a rectangle that is 8 by 5.

- A. True
- B. False

SET 59 • 6.G.A.1

# FORWARD 2

Use the Pythagorean Theorem to find a missing height • Grade 8 • Geometry •  
8.G.B.7

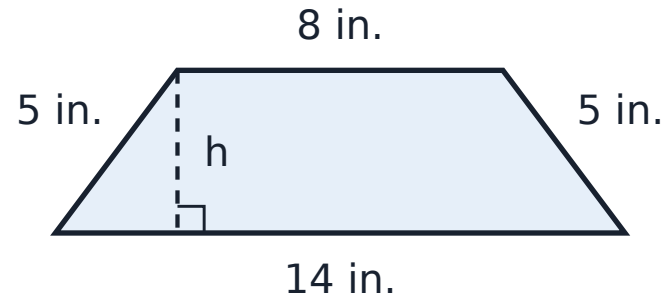
A parallelogram has a slanted side of 5 cm. The slanted side, the height, and a 3 cm piece of the base form a right triangle.

What is the height?

Height: \_\_\_\_\_

Find the height  $h$  of the trapezoid.

$h =$  \_\_\_\_\_



A ramp forms a right triangle. The ramp is 13 ft long and its base is 12 ft long.

How tall is the ramp?

**A.** 1 ft

**B.** 5 ft

**C.** 17.7 ft

**D.** 25 ft

A parallelogram has a slanted side of 10 m. The slanted side, the height, and a 6 m piece of the base form a right triangle.

What is the height?

Height: \_\_\_\_\_

## True or false?

A trapezoid has a slanted side of 10 in. The slanted side, the height, and a 6 in. piece of the base form a right triangle. The height is 8 in.

**A.** True

**B.** False

# Area of polygons by composing and decomposing

|                   |                                                  |                   |
|-------------------|--------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 6 main question                            | <b>6.G.A.1</b>    |
| <b>BACKWARD 1</b> | Area of rectilinear figures                      | <b>3.MD.C.7.d</b> |
| <b>BACKWARD 2</b> | Missing side lengths of rectangles               | <b>4.MD.A.3</b>   |
| <b>BACKWARD 3</b> | Area of triangles                                | <b>6.G.A.1</b>    |
| <b>FORWARD 1</b>  | Area of composite figures in real-world problems | <b>7.G.B.6</b>    |
| <b>FORWARD 2</b>  | Volume of composite solids                       | <b>8.G.C.9</b>    |



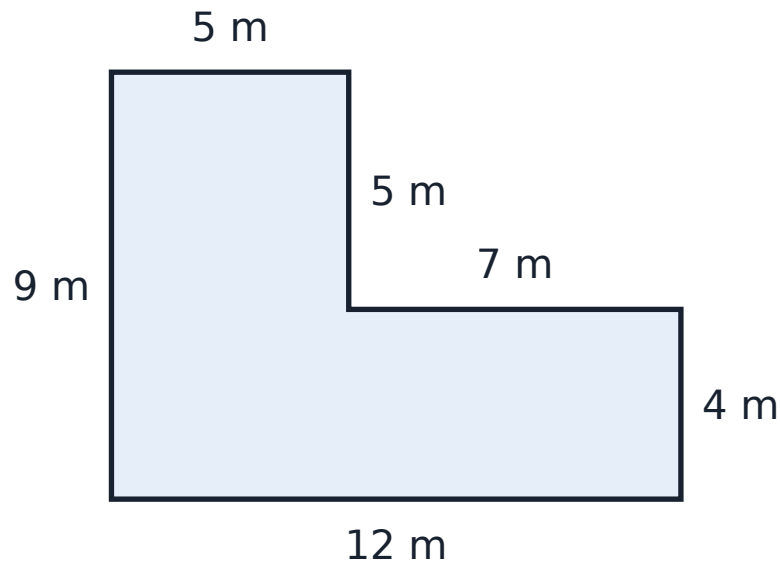
SET 60 • 6.G.A.1

# MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.1

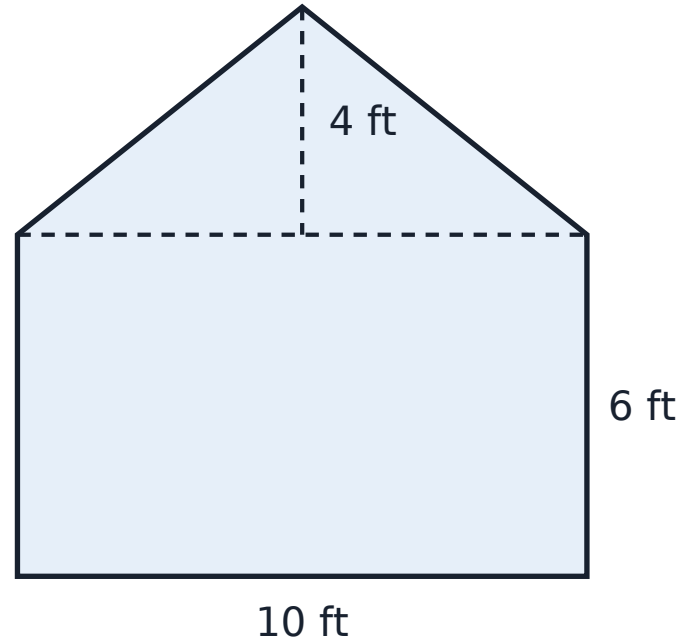
Find the area of the figure.

Area: \_\_\_\_\_



Find the area of the figure.

Area: \_\_\_\_\_



A rectangular yard is 20 m by 15 m. A triangular garden in the yard has a base of 6 m and a height of 5 m.

What is the area of the yard that is NOT garden?

Area: \_\_\_\_\_

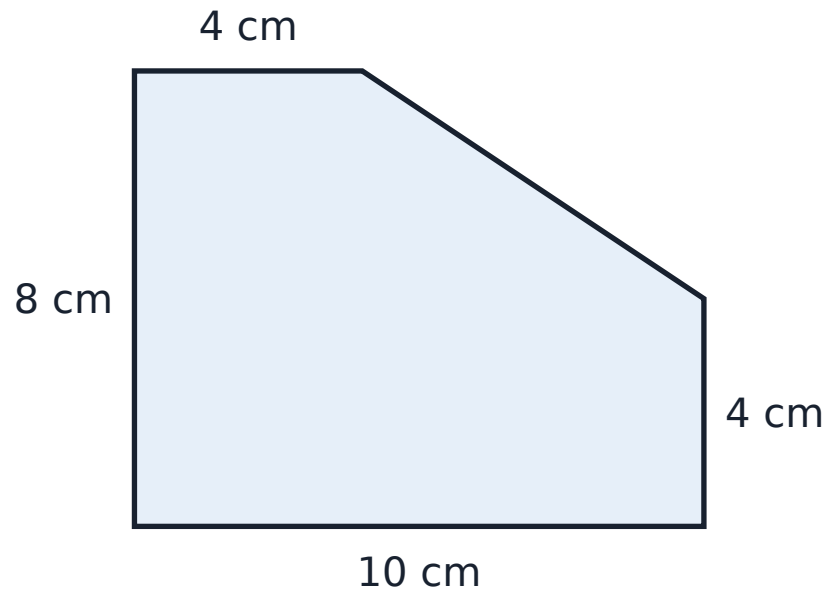
A polygon is split into a rectangle that is 8 by 3 and two triangles. Each triangle has a base of 2 and a height of 3.

What is the area of the polygon?

- A.** 24 square units
- B.** 27 square units
- C.** 30 square units
- D.** 36 square units

Find the area of the figure.

Area: \_\_\_\_\_



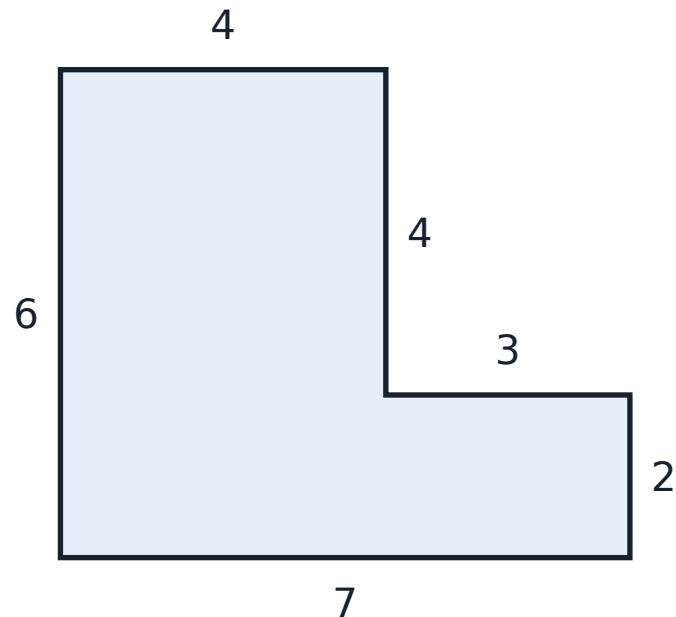
SET 60 • 6.G.A.1

# BACKWARD 1

Area of rectilinear figures • Grade 3 • Measurement & Data • 3.MD.C.7.d

Find the area of the figure.

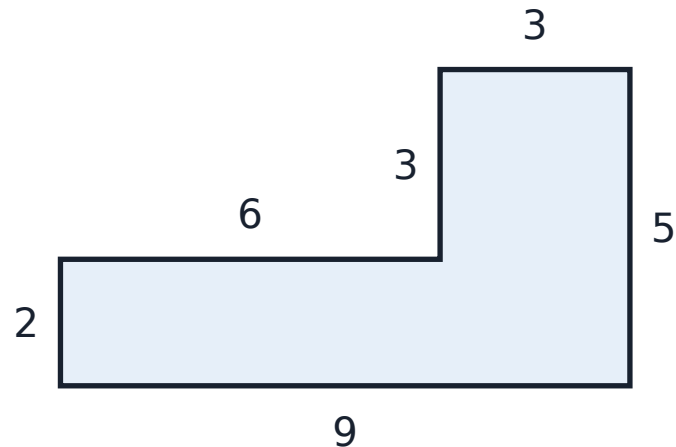
Area: \_\_\_\_\_





Find the area of the figure.

Area: \_\_\_\_\_



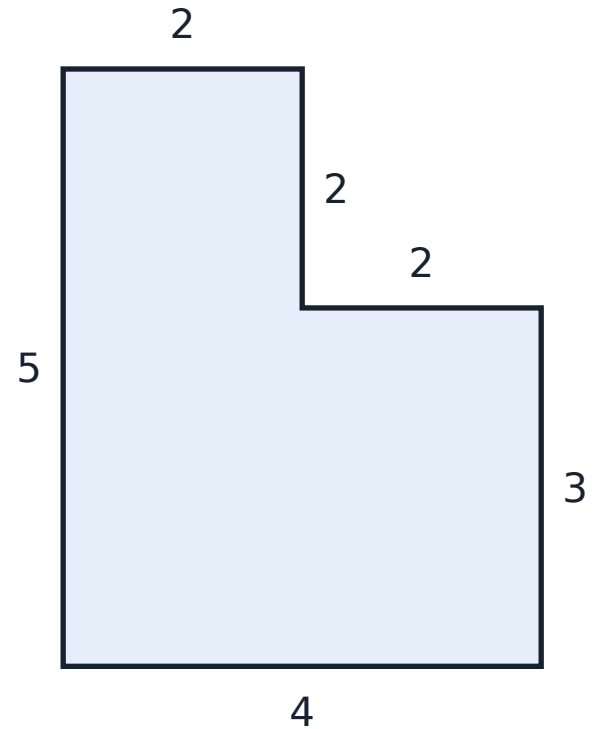
## True or false?

A figure made of a 5-by-2 rectangle and a 3-by-4 rectangle has an area of 22 square units.

- A. True
- B. False

Which expression gives the area of the figure?

- A.**  $4 \times 5$
- B.**  $4 + 3 + 2 + 2$
- C.**  $4 \times 3 \times 2$
- D.**  $4 \times 3 + 2 \times 2$



A figure is made of two rectangles: 6 by 3 and 2 by 4.  
What is its area?

Area: \_\_\_\_\_

SET 60 • 6.G.A.1

# BACKWARD 2

Missing side lengths of rectangles • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle has an area of 48 square feet and a width of 6 feet.

What is its length?

Length: \_\_\_\_\_

A rectangle has a perimeter of 30 cm and a length of 9 cm.

What is its width?

Width: \_\_\_\_\_

## True or false?

A rectangle with an area of 36 square cm and a length of 9 cm has a width of 4 cm.

**A.** True

**B.** False



A rectangle has an area of 56 square inches and a width of 7 inches.

What is its length?

**A.** 8 inches

**B.** 9 inches

**C.** 49 inches

**D.** 63 inches

The rectangle has an area of 60 square meters.

What is the missing side length?



Length: \_\_\_\_\_

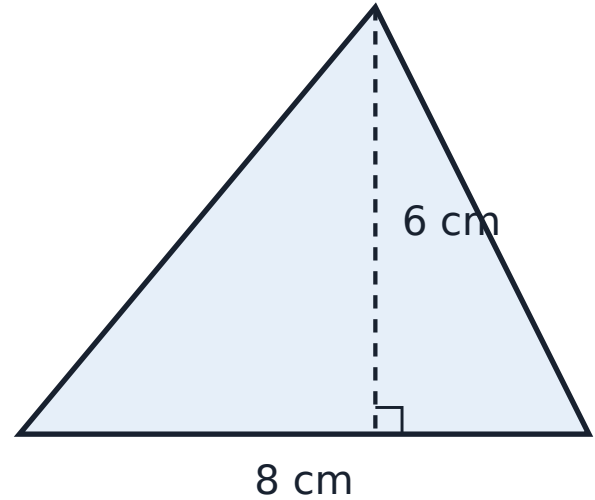
SET 60 • 6.G.A.1

# BACKWARD 3

Area of triangles • Grade 6 • Geometry • 6.G.A.1

Find the area of the triangle.

Area: \_\_\_\_\_



A triangle has a base of 10 m and a height of 3 m.

What is its area?

Area: \_\_\_\_\_

## True or false?

A right triangle with legs of 4 in. and 6 in. has an area of 24 square inches.

**A.** True

**B.** False

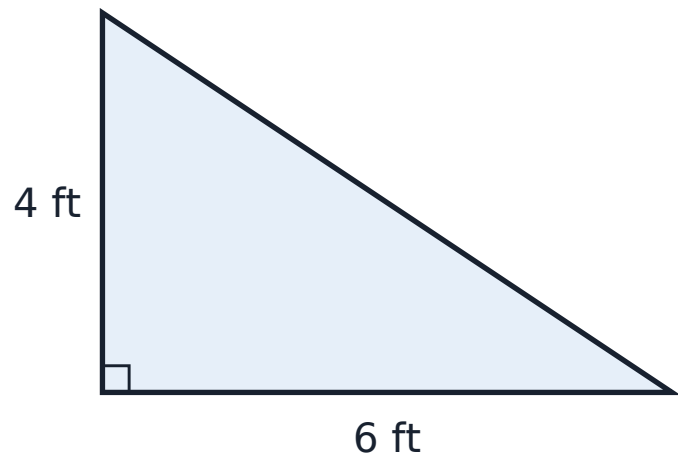
A triangle has a base of 5 ft and a height of 8 ft.

What is its area?

- A.** 13 square feet
- B.** 20 square feet
- C.** 26 square feet
- D.** 40 square feet

Find the area of the triangle.

Area: \_\_\_\_\_





SET 60 • 6.G.A.1

# FORWARD 1

Area of composite figures in real-world problems • Grade 7 • Geometry •  
7.G.B.6

A deck is made of a rectangle that is 12 ft by 10 ft and a triangle with a base of 10 ft and a height of 6 ft.

What is the area of the deck?

Area: \_\_\_\_\_

A trapezoid-shaped garden has bases of 8 m and 14 m and a height of 5 m. It has a square pond with 2 m sides.

What is the area of the garden without the pond?

Area: \_\_\_\_\_

A regular hexagon is split into 6 equal triangles. Each triangle has a base of 4 cm and a height of 3.5 cm.

What is the area of the hexagon?

- A.** 21 square cm
- B.** 24 square cm
- C.** 42 square cm
- D.** 84 square cm

An L-shaped prism has a base area of 30 square feet and a height of 4 feet.

What is its volume?

Volume: \_\_\_\_\_

## True or false?

A figure made of a 5-by-5 square and a triangle with a base of 5 and a height of 2 has an area of 30 square units.

- A.** True
- B.** False

SET 60 • 6.G.A.1

# FORWARD 2

Volume of composite solids • Grade 8 • Geometry • 8.G.C.9

A silo is a cylinder with a radius of 3 m and a height of 10 m, topped by a hemisphere with a radius of 3 m.

What is its volume in terms of  $\pi$ ?

Volume: \_\_\_\_\_



A cone has a radius of 3 in. and a height of 4 in. A hemisphere with a radius of 3 in. sits on top.

What is the total volume in terms of  $\pi$ ?

Volume: \_\_\_\_\_

A cone with a radius of 2 and a height of 3 is cut out of a cylinder with a radius of 2 and a height of 6.

What volume is left?

**A.**  $20\pi$

**B.**  $28\pi$

**C.**  $24\pi$

**D.**  $4\pi$

A cylinder and a cone each have a radius of 1 and a height of 2.

How much more volume does the cylinder have, in terms of  $\pi$ ?

Difference: \_\_\_\_\_

## True or false?

A sphere with a radius of 3 has the same volume as a cylinder with a radius of 3 and a height of 4.

**A.** True

**B.** False

# Find volume by packing a prism with unit-fraction cubes

|                   |                                         |                   |
|-------------------|-----------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 6 main question                   | <b>6.G.A.2</b>    |
| <b>BACKWARD 1</b> | Volume by counting unit cubes           | <b>5.MD.C.4</b>   |
| <b>BACKWARD 2</b> | The volume of one small cube            | <b>5.NF.B.4.a</b> |
| <b>BACKWARD 3</b> | Volume as the number of unit cubes      | <b>5.MD.C.3.b</b> |
| <b>FORWARD 1</b>  | Volume of prisms in real-world problems | <b>7.G.B.6</b>    |
| <b>FORWARD 2</b>  | Volume of cylinders, cones, and spheres | <b>8.G.C.9</b>    |

SET 61 • 6.G.A.2

# MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.2

How many cubes with  $\frac{1}{2}$ -inch edges fill a box that is 2 in. by  $1\frac{1}{2}$  in. by 1 in.?

Cubes: \_\_\_\_\_

A box is 3 in. by 2 in. by  $1\frac{1}{2}$  in.

How many cubes with  $\frac{1}{2}$ -inch edges fill the box?

Cubes: \_\_\_\_\_



## True or false?

A prism is packed with 48 cubes that each have  $\frac{1}{2}$ -unit edges. Its volume is 6 cubic units.

- A.** True
- B.** False

A prism is packed with 64 cubes that each have  $\frac{1}{4}$ -inch edges.

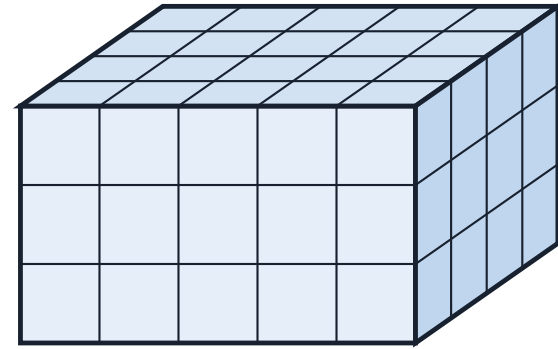
What is the volume of the prism?

- A.**  $\frac{1}{4}$  cubic inch
- B.** 1 cubic inch
- C.** 4 cubic inches
- D.** 16 cubic inches

The prism is packed with cubes that have  $\frac{1}{2}$ -cm edges.

It is 5 cubes long, 4 cubes wide, and 3 cubes tall.

What is its volume?



Volume: \_\_\_\_\_

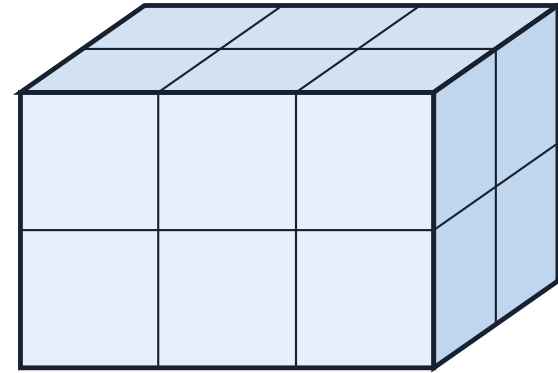
SET 61 • 6.G.A.2

# BACKWARD 1

Volume by counting unit cubes • Grade 5 • Measurement & Data • 5.MD.C.4

Each cube is 1 cubic unit.

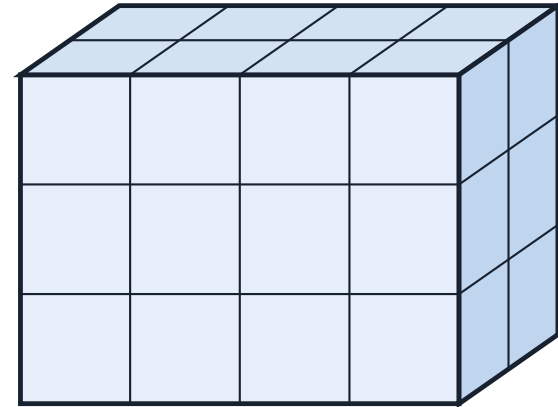
What is the volume of the prism?



Volume: \_\_\_\_\_

Each cube is 1 cubic unit.

What is the volume of the prism?



Volume: \_\_\_\_\_

## True or false?

A prism made of 2 layers with 6 unit cubes in each layer has a volume of 12 cubic units.

- A. True
- B. False

A box holds 3 layers with 5 unit cubes in each layer.

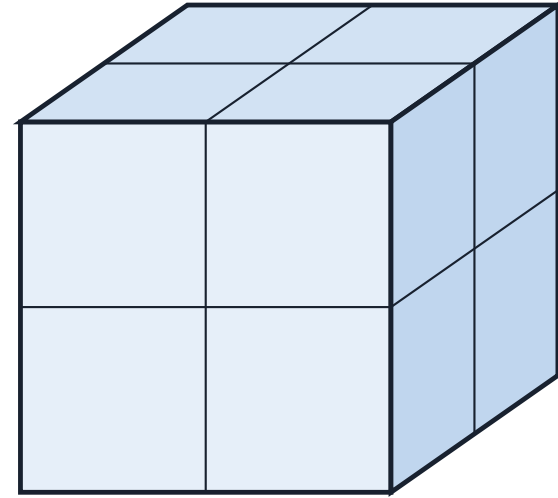
What is its volume?

- A.** 8 cubic units
- B.** 15 cubic units
- C.** 35 cubic units
- D.** 53 cubic units



Each cube is 1 cubic unit.

What is the volume of the prism?



Volume: \_\_\_\_\_

SET 61 • 6.G.A.2

# BACKWARD 2

The volume of one small cube • Grade 5 • Number & Operations–Fractions •  
5.NF.B.4.a

Multiply.

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

Product: \_\_\_\_\_

Multiply.

$$\frac{1}{4} \times \frac{1}{4} \times \frac{1}{4}$$

Product: \_\_\_\_\_

## True or false?

$$\frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$$

**A.** True

**B.** False

Multiply.

$$24 \times \frac{1}{8}$$

**A.**  $\frac{1}{3}$

**B.** 3

**C.** 32

**D.** 192

Multiply.

$$64 \times \frac{1}{64}$$

Product: \_\_\_\_\_

SET 61 • 6.G.A.2

# BACKWARD 3

Volume as the number of unit cubes • Grade 5 • Measurement & Data •  
5.MD.C.3.b



## True or false?

A solid that can be packed with 20 unit cubes with no gaps has a volume of 20 cubic units.

- A. True
- B. False

A box is filled with 36 unit cubes with no gaps or overlaps.

What is its volume?

- A.** 6 cubic units
- B.** 12 cubic units
- C.** 36 cubic units
- D.** 36 square units

A box is filled with 45 unit cubes with no gaps.  
What is its volume?

Volume: \_\_\_\_\_

## True or false?

Volume is measured in square units.

**A.** True

**B.** False

Which unit could be used to measure volume?

- A.** Square centimeters
- B.** Cubic centimeters
- C.** Centimeters
- D.** Kilograms

SET 61 • 6.G.A.2

# FORWARD 1

Volume of prisms in real-world problems • Grade 7 • Geometry • 7.G.B.6

A triangular prism has triangle bases with a base of 5 in. and a height of 4 in. The prism is 8 in. long.

What is its volume?

Volume: \_\_\_\_\_

A prism has a base area of  $12\frac{1}{2}$  square cm and a height of 4 cm.

What is its volume?

Volume: \_\_\_\_\_



A cube has edges of 2.5 cm.

What is its volume?

- A.** 6.25 cubic cm
- B.** 7.5 cubic cm
- C.** 15 cubic cm
- D.** 15.625 cubic cm

An aquarium is 30 in. long, 12 in. wide, and 16 in. tall. It is filled with water to a height of 12 in.

What is the volume of the water?

Volume: \_\_\_\_\_

## True or false?

A triangular prism with a base area of 15 square units and a length of 6 units has a volume of 90 cubic units.

**A.** True

**B.** False

SET 61 • 6.G.A.2

# FORWARD 2

Volume of cylinders, cones, and spheres • Grade 8 • Geometry • 8.G.C.9

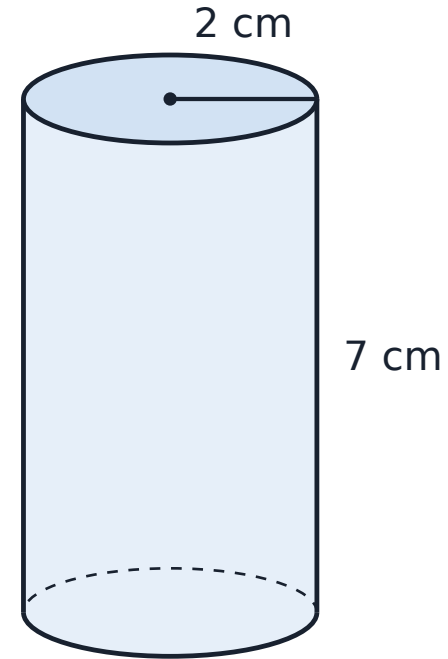
A cylinder has a radius of 5 and a height of 4.  
What is its volume in terms of  $\pi$ ?

Volume: \_\_\_\_\_

Find the volume of the cylinder.

Use 3.14 for  $\pi$ .

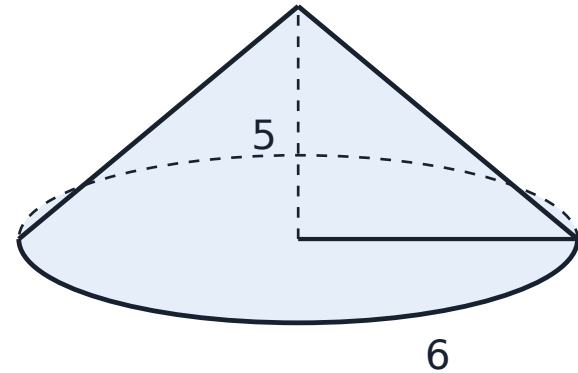
Volume: \_\_\_\_\_



A cone has a radius of 6 and a height of 5.

What is its volume?

- A.**  $150\pi$                       **B.**  $180\pi$   
**C.**  $30\pi$                         **D.**  $60\pi$



A sphere has a radius of 6.

What is its volume in terms of  $\pi$ ?

Volume: \_\_\_\_\_



## True or false?

A cylinder has a greater volume than a cone with the same radius and height.

- A.** True
- B.** False

# Apply $V = l w h$ and $V = b h$ with fractional edge lengths

## MAIN

Grade 6 main question

**6.G.A.2**

## BACKWARD 1

Volume formula with whole numbers

**5.MD.C.5.b**

## BACKWARD 2

Multiplying mixed numbers

**5.NF.B.6**

## BACKWARD 3

Multiplying fractions

**5.NF.B.4.a**

## FORWARD 1

Volume of triangular and rectangular prisms

**7.G.B.6**

## FORWARD 2

Volume of cylinders using  $V = Bh$

**8.G.C.9**

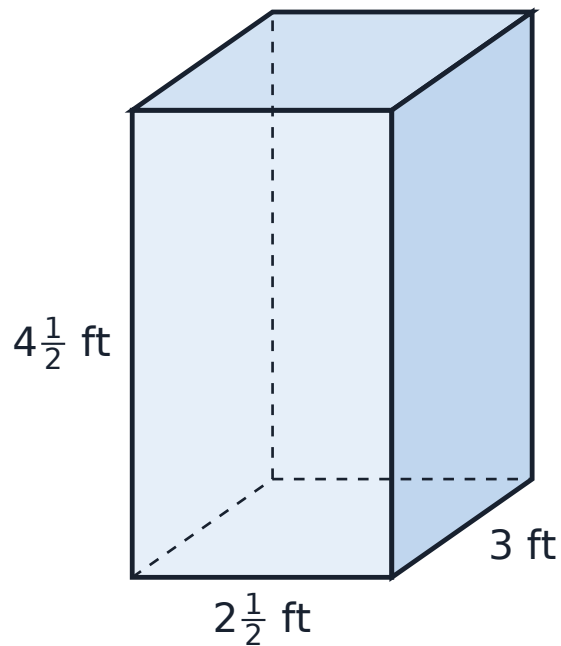
SET 62 • 6.G.A.2

# MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.2

Find the volume of the rectangular prism.

Volume: \_\_\_\_\_



A box is  $1\frac{1}{2}$  ft long, 2 ft wide, and  $\frac{3}{4}$  ft tall.

What is its volume?

- A.**  $2\frac{1}{4}$  cubic feet
- B.**  $2\frac{3}{4}$  cubic feet
- C.** 3 cubic feet
- D.**  $4\frac{1}{4}$  cubic feet

Use  $V = l \times w \times h$  to find the volume of a prism with  $l = 5$  cm,  $w = 2\frac{1}{2}$  cm, and  $h = 1\frac{1}{5}$  cm.

$V =$  \_\_\_\_\_

A cube has edges of  $1\frac{1}{2}$  feet.

What is its volume?

Volume: \_\_\_\_\_

A prism has a base area of  $4\frac{1}{2}$  square inches and a height of  $2\frac{2}{3}$  inches.

Use  $V = b \times h$  to find its volume.

$V =$  \_\_\_\_\_



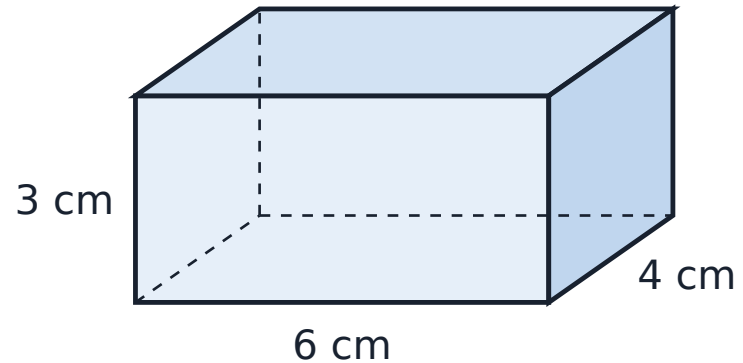
SET 62 • 6.G.A.2

# BACKWARD 1

Volume formula with whole numbers • Grade 5 • Measurement & Data •  
5.MD.C.5.b

Find the volume of the prism.

Volume: \_\_\_\_\_



A prism has a length of 10 m, a width of 2 m, and a height of 5 m.

What is its volume?

Volume: \_\_\_\_\_

## True or false?

A cube with 3-inch edges has a volume of 27 cubic inches.

**A.** True

**B.** False

A box is 8 in. by 5 in. by 2 in.

What is its volume?

- A.** 15 cubic inches
- B.** 40 cubic inches
- C.** 80 cubic inches
- D.** 160 cubic inches

A prism has a base area of 20 square cm and a height of 6 cm.

What is its volume?

Volume: \_\_\_\_\_

SET 62 • 6.G.A.2

# BACKWARD 2

Multiplying mixed numbers • Grade 5 • Number & Operations–Fractions •  
5.NF.B.6

Multiply.

$$2\frac{1}{2} \times 3$$

Product: \_\_\_\_\_



Multiply.

$$1\frac{1}{2} \times \frac{3}{4}$$

Product: \_\_\_\_\_

**True or false?**

$$2\frac{1}{2} \times 2 = 5$$

**A.** True

**B.** False

$$3 \times 4\frac{1}{2} = ?$$

**A.**  $7\frac{1}{2}$

**B.**  $12\frac{1}{6}$

**C.**  $12\frac{1}{2}$

**D.**  $13\frac{1}{2}$

Multiply.

$$1\frac{1}{3} \times 1\frac{1}{2}$$

Product: \_\_\_\_\_

SET 62 • 6.G.A.2

# BACKWARD 3

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{3}{4} \times \frac{2}{3}$$

Product: \_\_\_\_\_

Multiply.

$$\frac{5}{2} \times \frac{9}{2}$$

Product: \_\_\_\_\_

## True or false?

$$\frac{3}{2} \times \frac{3}{2} = \frac{9}{4}$$

**A.** True

**B.** False



$$\frac{1}{2} \times \frac{4}{5} = ?$$

**A.**  $\frac{2}{5}$

**B.**  $\frac{4}{7}$

**C.**  $\frac{5}{7}$

**D.**  $\frac{8}{5}$

Multiply.

$$\frac{6}{5} \times 5$$

Product: \_\_\_\_\_

SET 62 • 6.G.A.2

# FORWARD 1

Volume of triangular and rectangular prisms • Grade 7 • Geometry • 7.G.B.6

A triangular prism has triangle bases with a base of 6 cm and a height of 4 cm. The prism is 9 cm long.

What is its volume?

Volume: \_\_\_\_\_

A tent is a triangular prism. Its triangle has a base of 2.4 m and a height of 1.5 m. The tent is 3 m long.

What is its volume?

Volume: \_\_\_\_\_

A prism has a base area of 7.5 square feet and a height of 2.4 feet.

What is its volume?

- A.** 9.9 cubic feet
- B.** 15 cubic feet
- C.** 18 cubic feet
- D.** 180 cubic feet

A box is 4.5 cm by 2 cm by 3.2 cm.

What is its volume?

Volume: \_\_\_\_\_

## True or false?

A triangular prism with a triangle area of 10 square inches and a length of 7 inches has a volume of 70 cubic inches.

**A.** True

**B.** False



SET 62 • 6.G.A.2

# FORWARD 2

Volume of cylinders using  $V = Bh$  • Grade 8 • Geometry • 8.G.C.9

A cylinder has a base area of  $12\pi$  square cm and a height of 5 cm.

What is its volume in terms of  $\pi$ ?

Volume: \_\_\_\_\_

A can has a radius of 3 in. and a height of 8 in.  
What is its volume in terms of  $\pi$ ?

Volume: \_\_\_\_\_

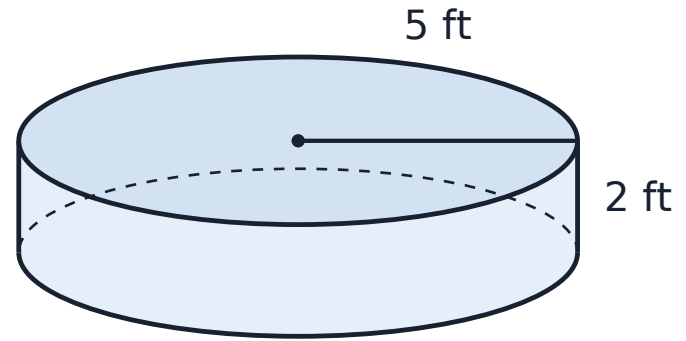
A cylinder has a radius of 1.5 m and a height of 4 m.

What is its volume?

- A.**  $12\pi$  cubic meters
- B.**  $9\pi$  cubic meters
- C.**  $3\pi$  cubic meters
- D.**  $6\pi$  cubic meters

Find the volume of the cylinder in terms of  $\pi$ .

Volume: \_\_\_\_\_



## True or false?

Doubling the height of a cylinder doubles its volume.

- A.** True
- B.** False

# Draw polygons in the coordinate plane from the coordinates of their vertices

|                   |                                               |                   |
|-------------------|-----------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 6 main question                         | <b>6.G.A.3</b>    |
| <b>BACKWARD 1</b> | Plotting points in all four quadrants         | <b>6.NS.C.6.c</b> |
| <b>BACKWARD 2</b> | Attributes of quadrilaterals                  | <b>3.G.A.1</b>    |
| <b>BACKWARD 3</b> | Naming shapes by sides and angles             | <b>2.G.A.1</b>    |
| <b>FORWARD 1</b>  | Draw shapes with given conditions             | <b>7.G.A.2</b>    |
| <b>FORWARD 2</b>  | Draw images of polygons under transformations | <b>8.G.A.3</b>    |

SET 63 • 6.G.A.3

# MAIN

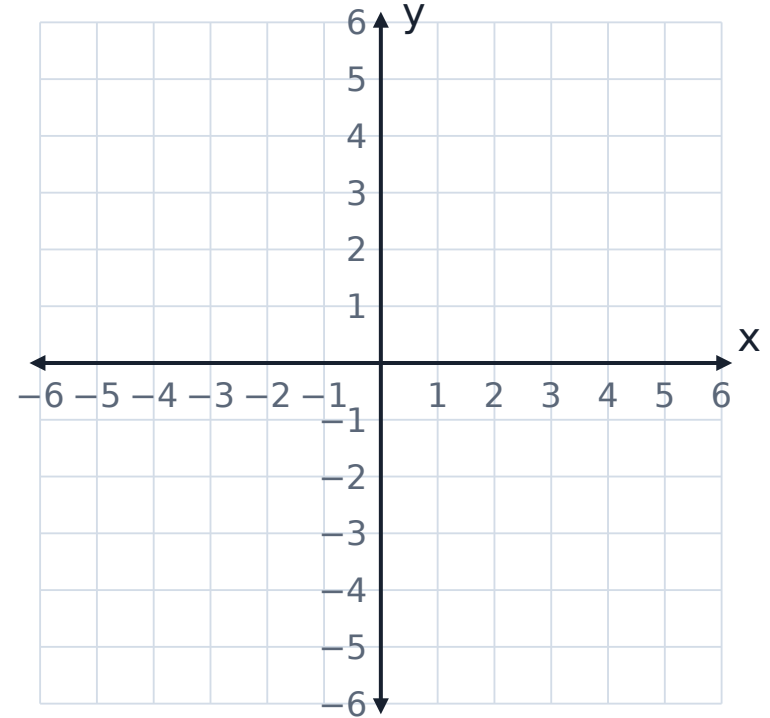
Grade 6 main question • Grade 6 • Geometry • 6.G.A.3



Plot  $A(-3, 2)$ ,  $B(4, 2)$ ,  $C(4, -3)$ , and  $D(-3, -3)$ . Connect them in order.

What polygon did you draw?

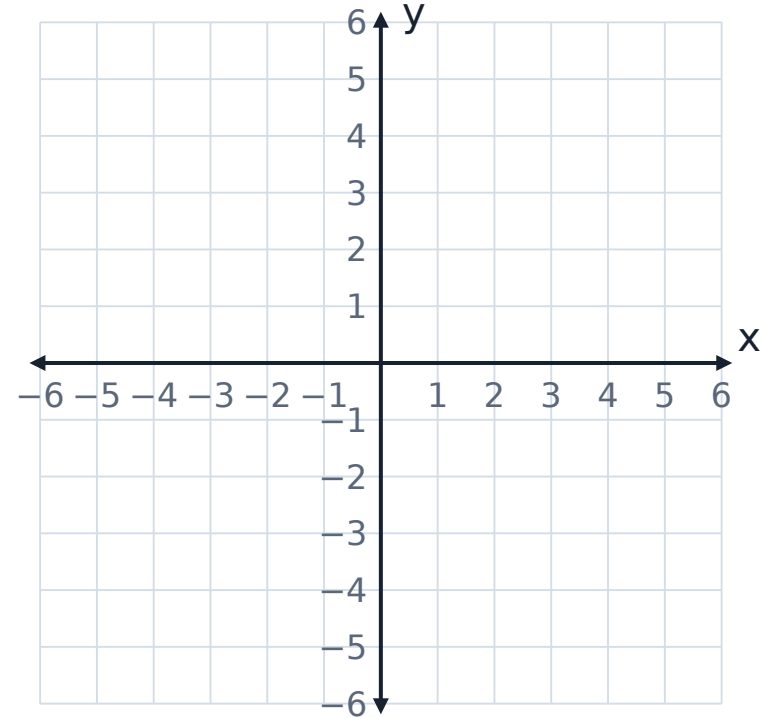
Polygon: \_\_\_\_\_



Plot  $P(0, 4)$ ,  $Q(3, -2)$ , and  $R(-3, -2)$ . Connect them in order.

What polygon did you draw?

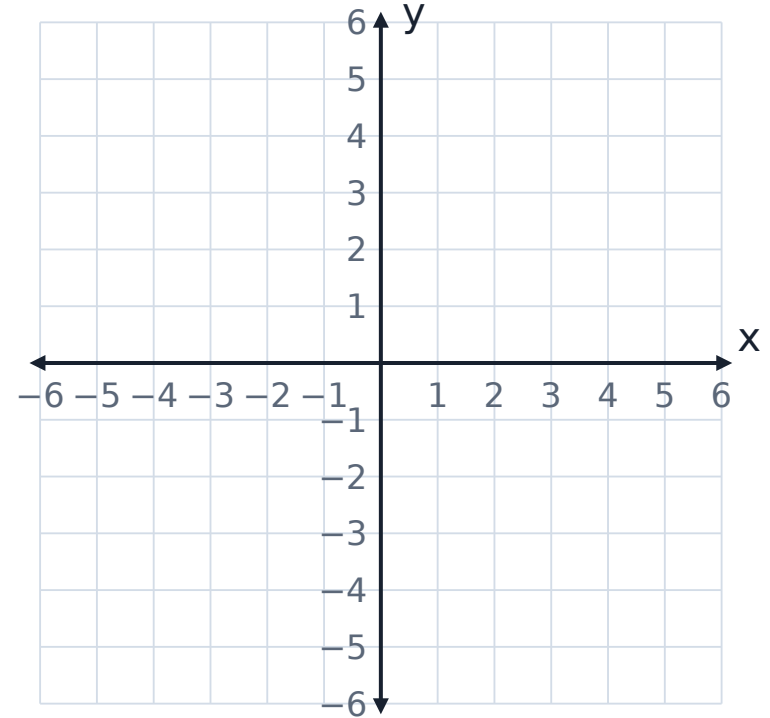
Polygon: \_\_\_\_\_



Plot  $E(-2, 1)$ ,  $F(2, 1)$ ,  $G(4, -3)$ , and  $H(-4, -3)$ . Connect them in order.

What polygon did you draw?

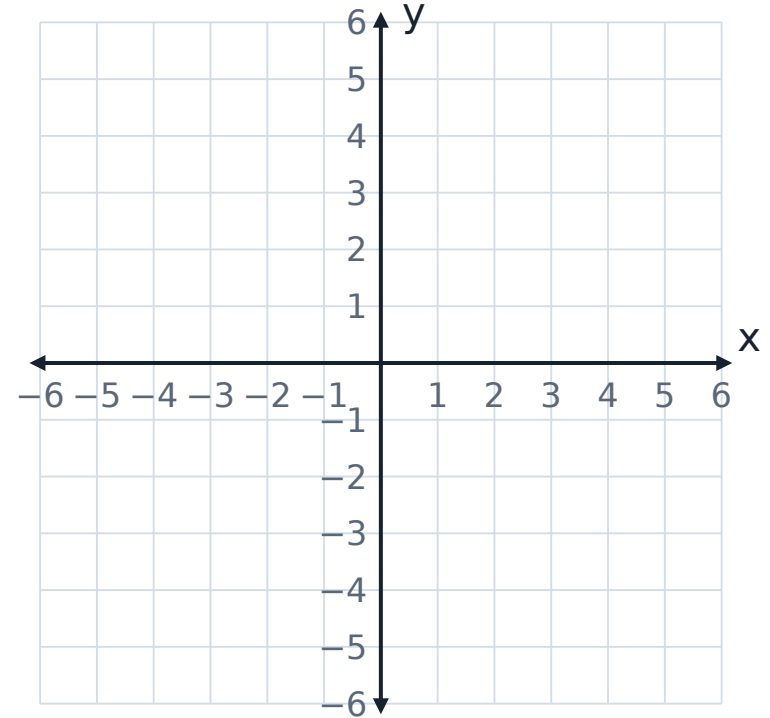
Polygon: \_\_\_\_\_



Plot  $J(-1, 3)$ ,  $K(3, 3)$ ,  $L(3, -1)$ ,  
and  $M(-1, -1)$ . Connect  
them in order.

What polygon did you draw?

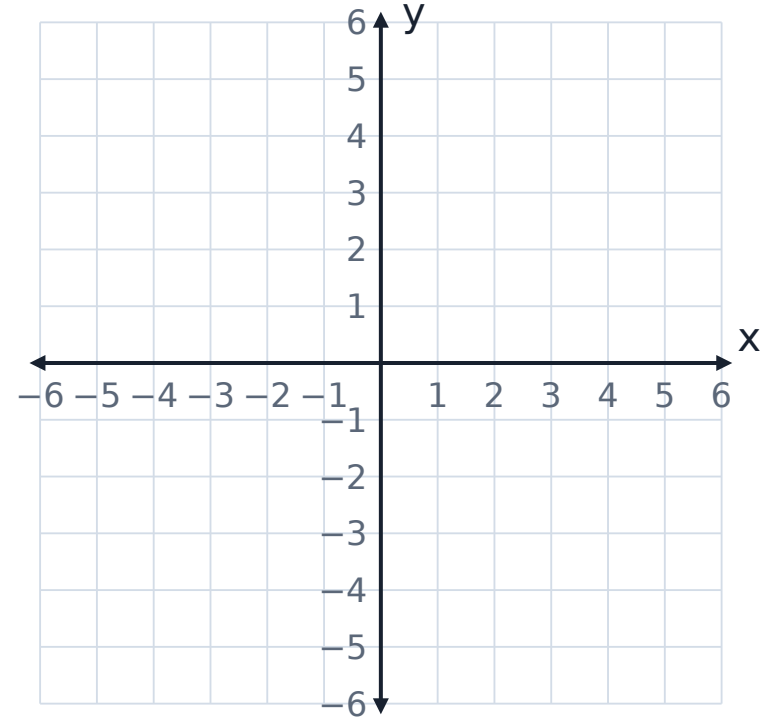
Polygon: \_\_\_\_\_



Plot  $W(-5, -2)$ ,  $X(-2, 3)$ ,  $Y(4, 3)$ , and  $Z(1, -2)$ . Connect them in order.

What polygon did you draw?

Polygon: \_\_\_\_\_

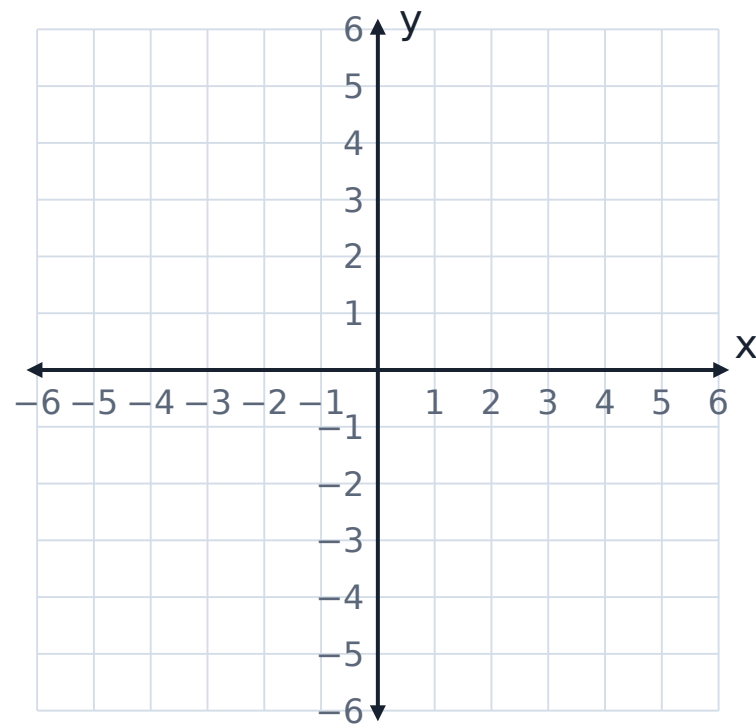


SET 63 • 6.G.A.3

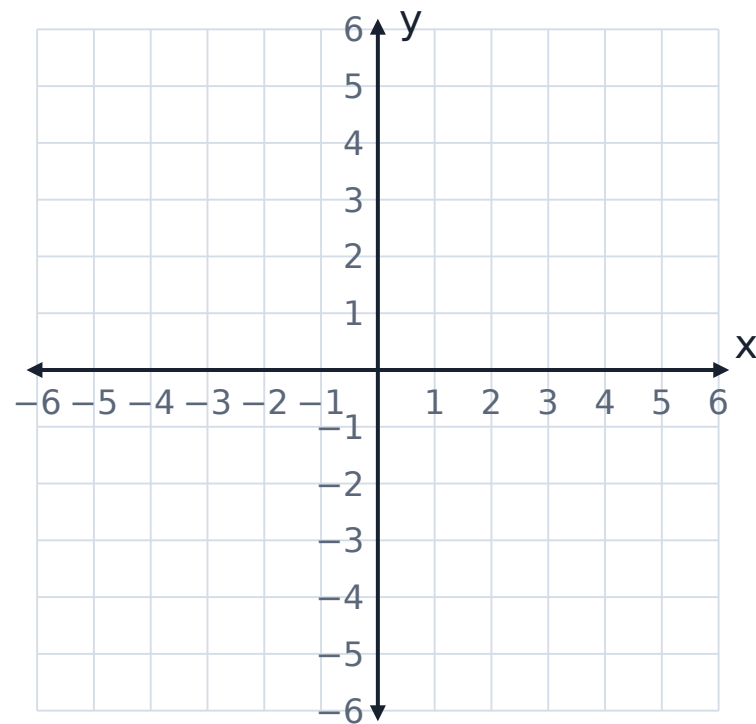
# BACKWARD 1

Plotting points in all four quadrants • Grade 6 • The Number System •  
6.NS.C.6.c

Plot and label point A( $-3, 2$ ).



Plot and label point B(4, -3).





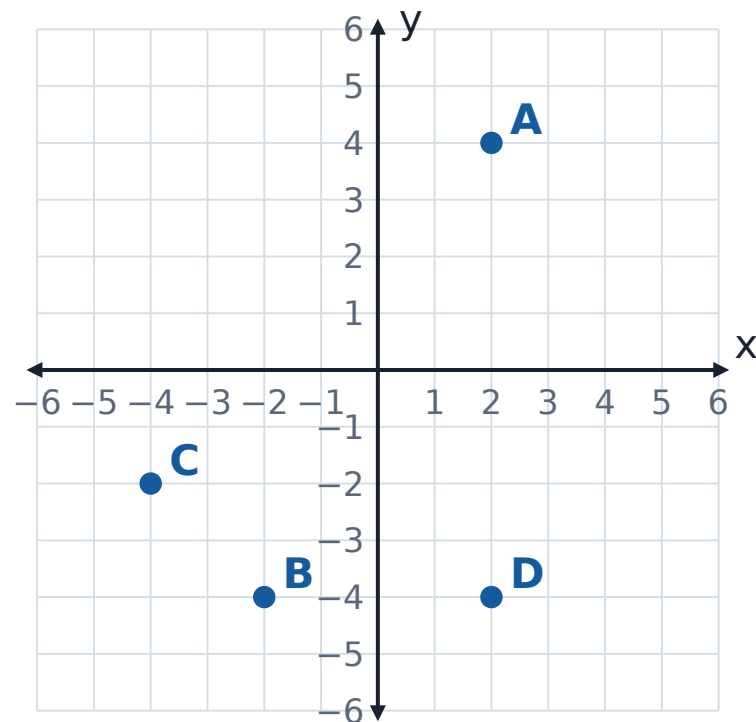
Which point is at  $(-2, -4)$ ?

**A.** Point A

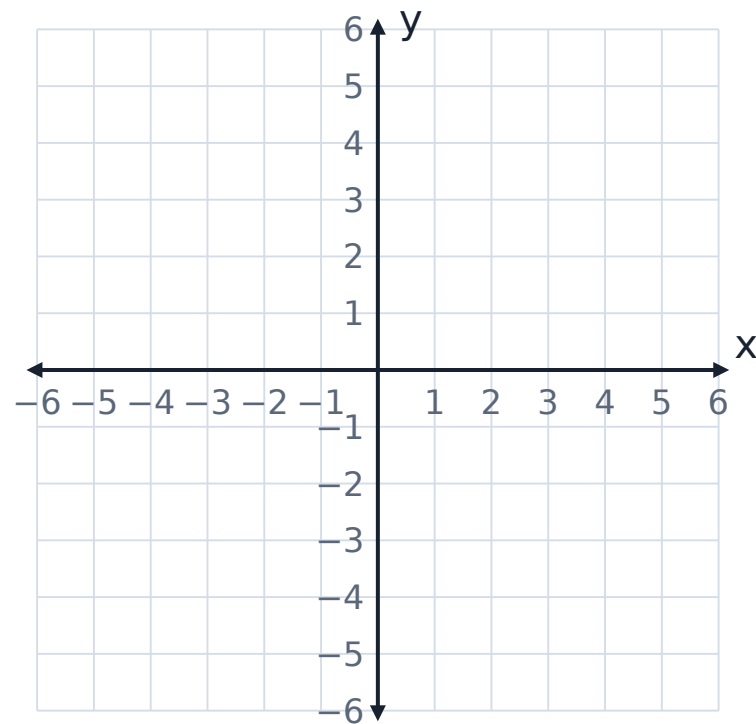
**B.** Point B

**C.** Point C

**D.** Point D



Plot and label point C(0, 4).

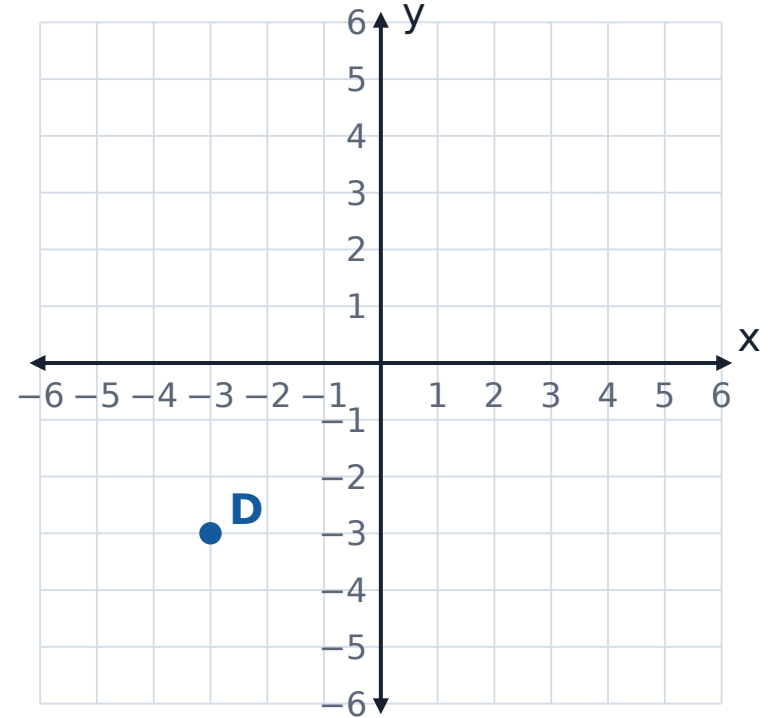


**True or false?**

Point D is at  $(-3, -3)$ .

**A.** True

**B.** False



SET 63 • 6.G.A.3

# BACKWARD 2

Attributes of quadrilaterals • Grade 3 • Geometry • 3.G.A.1

## True or false?

A rectangle has 4 right angles.

**A.** True

**B.** False

## True or false?

Opposite sides of a rectangle have the same length.

**A.** True

**B.** False

Which shape always has 4 sides of equal length and 4 right angles?

- A.** Square
- B.** Rectangle
- C.** Trapezoid
- D.** Rhombus

## True or false?

Every quadrilateral has 4 right angles.

**A.** True

**B.** False



How many sides does a quadrilateral have?

Sides: \_\_\_\_\_

SET 63 • 6.G.A.3

# BACKWARD 3

Naming shapes by sides and angles • Grade 2 • Geometry • 2.G.A.1

A shape has 3 sides and 3 angles.

What is it called?

Shape: \_\_\_\_\_

How many sides does a pentagon have?

**A.** 4

**B.** 5

**C.** 6

**D.** 8

## True or false?

A hexagon has 6 sides.

**A.** True

**B.** False

A shape has 4 sides and 4 angles.

What is it called?

Shape: \_\_\_\_\_

Which shape has the most sides?

- A.** Pentagon
- B.** Quadrilateral
- C.** Triangle
- D.** Hexagon

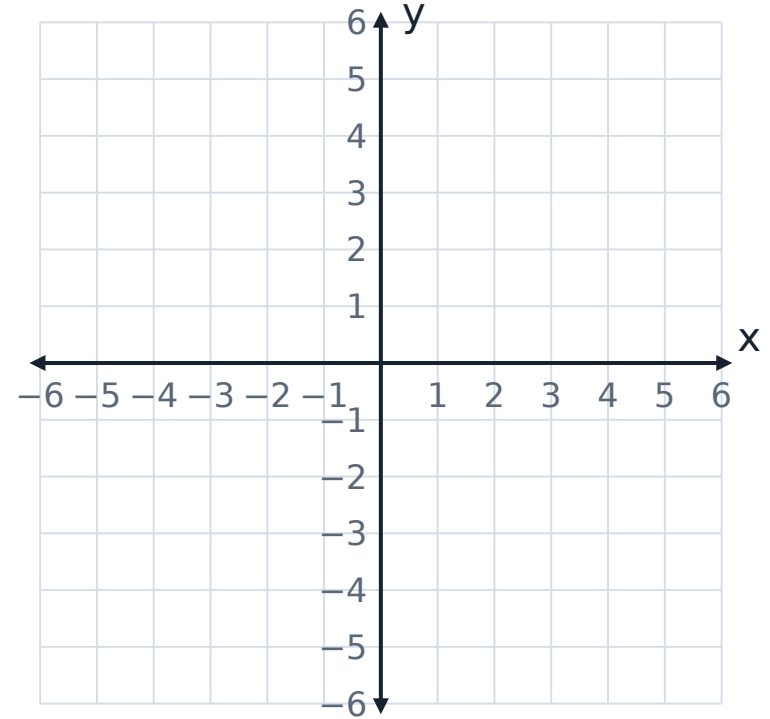
SET 63 • 6.G.A.3

# FORWARD 1

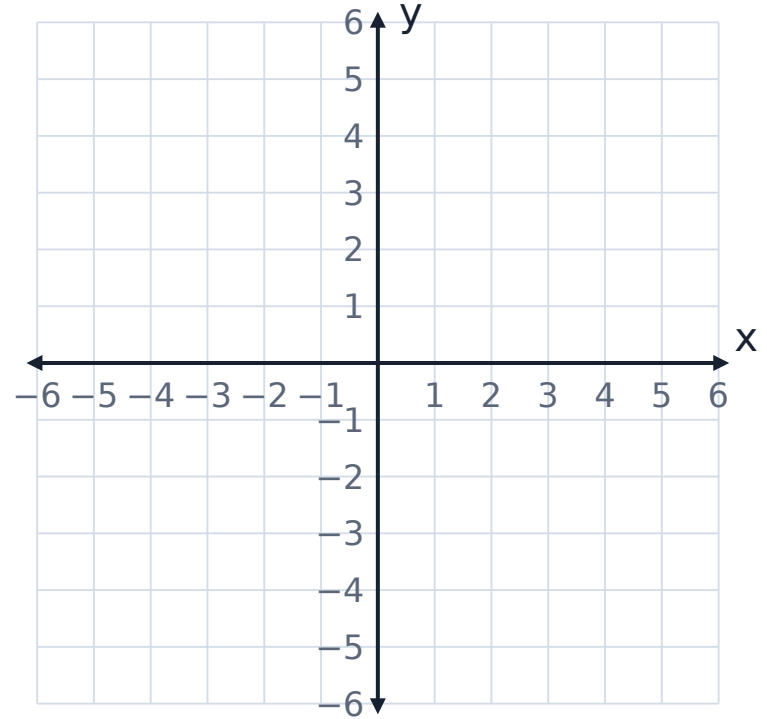
Draw shapes with given conditions • Grade 7 • Geometry • 7.G.A.2



Draw a rectangle with an area of 12 square units. One vertex is at the origin.



Draw a right triangle with legs of 4 units and 3 units. One vertex is at  $(-2, -2)$ .



Can a triangle have side lengths of 2 cm, 3 cm, and 6 cm? Write yes or no.

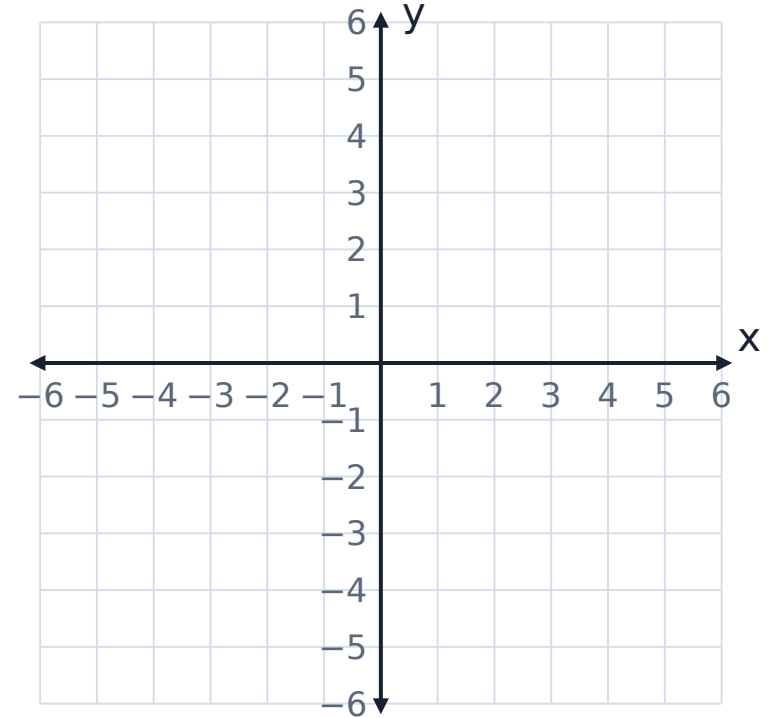
Answer: \_\_\_\_\_

A triangle has angles of  $90^\circ$  and  $45^\circ$ .

What is the third angle?

- A.**  $45^\circ$
- B.**  $55^\circ$
- C.**  $90^\circ$
- D.**  $135^\circ$

Draw a square with a side length of 5 units. One vertex is at  $(-3, 1)$ .



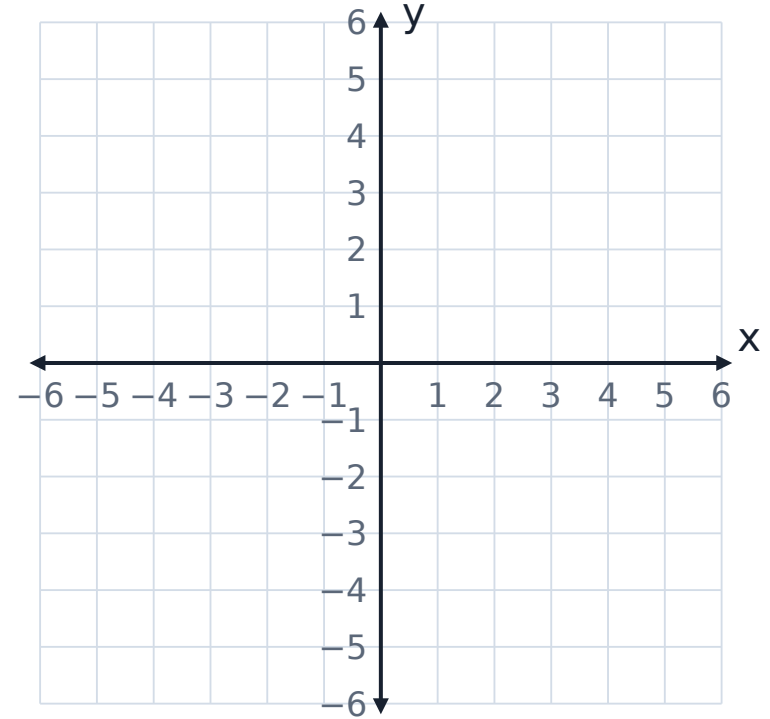
SET 63 • 6.G.A.3

# FORWARD 2

Draw images of polygons under transformations • Grade 8 • Geometry •  
8.G.A.3

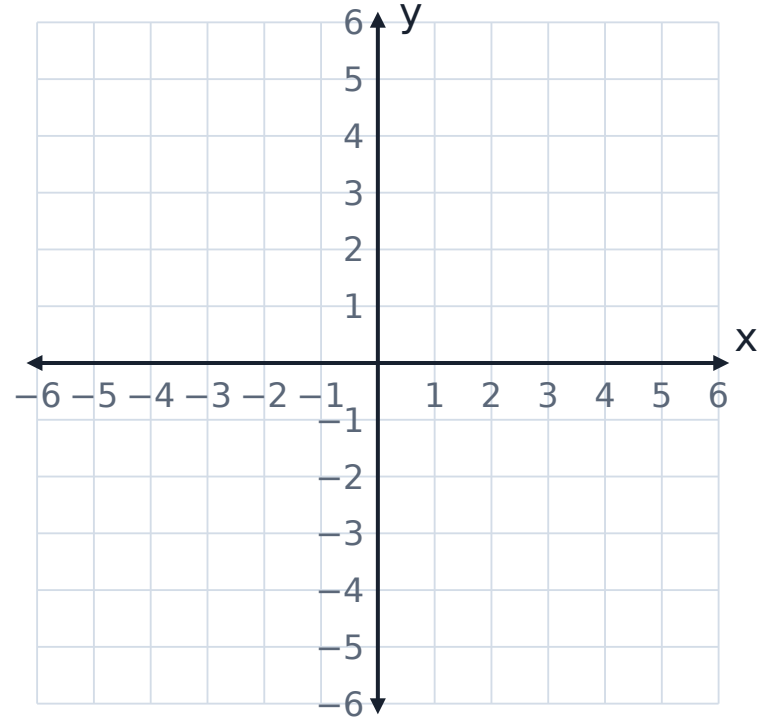
Triangle ABC has vertices  
 $A(1, 1)$ ,  $B(4, 1)$ , and  $C(1, 3)$ .

Draw its image after a  
translation 5 units left and 2  
units down.



Rectangle PQRS has vertices  $P(1, 2)$ ,  $Q(4, 2)$ ,  $R(4, 4)$ , and  $S(1, 4)$ .

Draw its image after a reflection across the x-axis.





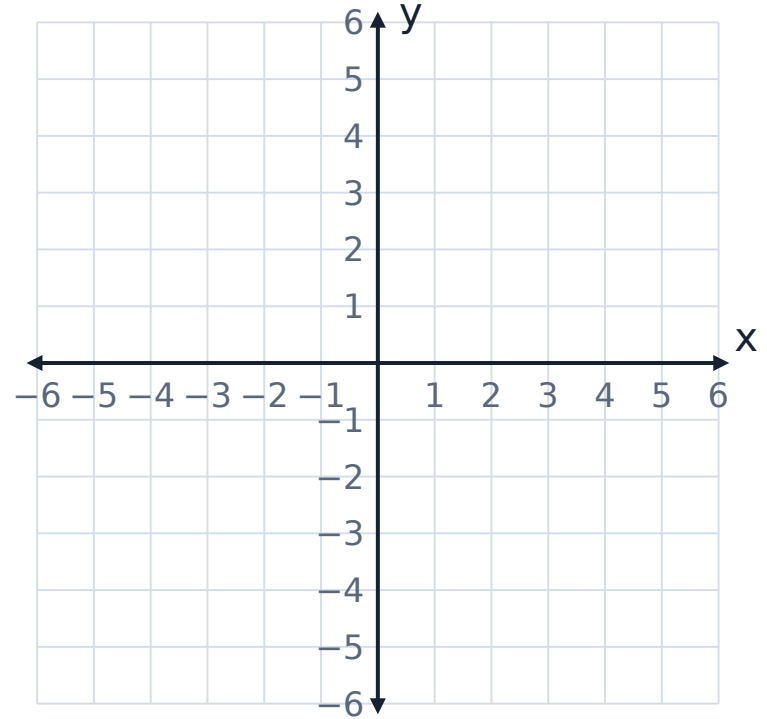
Triangle ABC has vertices A(2, 1), B(4, 3), and C(2, 3). It is rotated  $180^\circ$  about the origin.

What are the coordinates of B'?

B' = \_\_\_\_\_

Square EFGH has vertices  $E(1, 1)$ ,  $F(2, 1)$ ,  $G(2, 2)$ , and  $H(1, 2)$ .

Draw its image after a dilation by a scale factor of 3 centered at the origin.



## True or false?

A rectangle with vertices  $(1, 1)$ ,  $(3, 1)$ ,  $(3, 2)$ ,  $(1, 2)$  is reflected across the  $y$ -axis. One image vertex is  $(-3, 2)$ .

**A.** True

**B.** False

# Use coordinates to find side lengths and perimeters of polygons

|                   |                                                        |                   |
|-------------------|--------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 6 main question                                  | <b>6.G.A.3</b>    |
| <b>BACKWARD 1</b> | Distance between points with the same x or y           | <b>6.NS.C.8</b>   |
| <b>BACKWARD 2</b> | Absolute value                                         | <b>6.NS.C.7.c</b> |
| <b>BACKWARD 3</b> | Perimeter of polygons                                  | <b>3.MD.D.8</b>   |
| <b>FORWARD 1</b>  | Lengths between points with rational coordinates       | <b>7.NS.A.1.c</b> |
| <b>FORWARD 2</b>  | Find slanted side lengths with the Pythagorean Theorem | <b>8.G.B.8</b>    |

SET 64 • 6.G.A.3

# MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.3

A rectangle has vertices at  $(-3, 2)$ ,  $(4, 2)$ ,  $(4, -5)$ , and  $(-3, -5)$ .

What is its perimeter?

Perimeter: \_\_\_\_\_

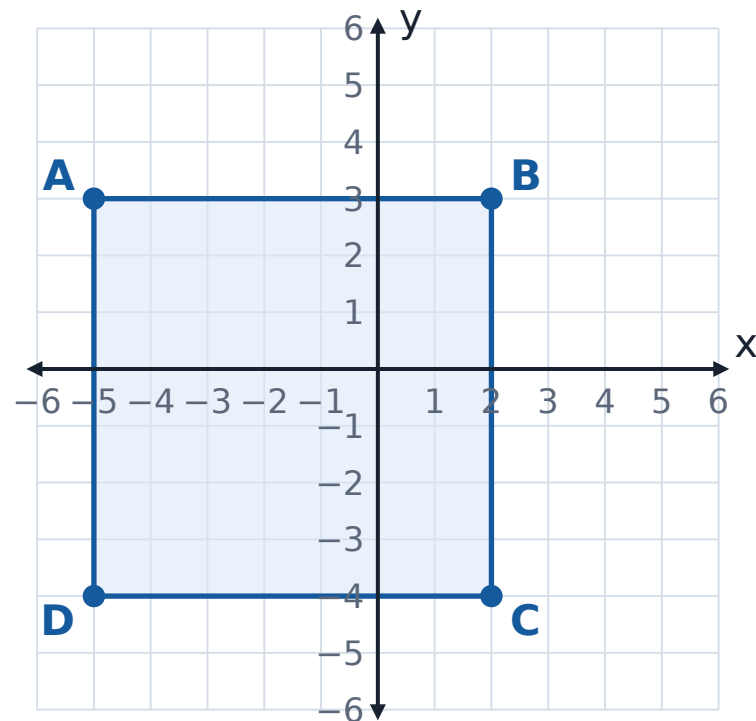
One side of a polygon goes from  $(-2, -3)$  to  $(-2, 5)$ .

How long is this side?

Length: \_\_\_\_\_

What is the length of side AB?

AB = \_\_\_\_\_





A triangle has vertices at  $(-4, 1)$ ,  $(2, 1)$ , and  $(2, -7)$ .

How long is the horizontal side?

- A.** 2 units
- B.** 6 units
- C.** 8 units
- D.** 10 units

A rectangle has vertices at  $(-5, -2)$ ,  $(1, -2)$ ,  $(1, 3)$ , and  $(-5, 3)$ .

What is its perimeter?

Perimeter: \_\_\_\_\_

SET 64 • 6.G.A.3

# BACKWARD 1

Distance between points with the same x or y • Grade 6 • The Number System  
• 6.NS.C.8

Find the distance between  $(-3, 2)$  and  $(4, 2)$ .

Distance: \_\_\_\_\_

Find the distance between  $(1, -4)$  and  $(1, 6)$ .

Distance: \_\_\_\_\_

## True or false?

The distance between  $(-5, 0)$  and  $(5, 0)$  is 10.

**A.** True

**B.** False

What is the distance between  $(2, -3)$  and  $(2, -8)$ ?

**A.**  $-11$

**B.**  $-5$

**C.**  $5$

**D.**  $11$

Find the distance between  $(-6, -1)$  and  $(-2, -1)$ .

Distance: \_\_\_\_\_



SET 64 • 6.G.A.3

# BACKWARD 2

Absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-5| + |2|$$

Value: \_\_\_\_\_

**True or false?**

$$|-7| = 7$$

**A.** True

**B.** False

$$|-3| + |4| = ?$$

**A.**  $-7$

**B.**  $-1$

**C.**  $1$

**D.**  $7$

Find the value.

$$|-6|$$

Value: \_\_\_\_\_

**True or false?**

$$|-2| + |-2| = 0$$

**A.** True

**B.** False

SET 64 • 6.G.A.3

# BACKWARD 3

Perimeter of polygons • Grade 3 • Measurement & Data • 3.MD.D.8

Find the perimeter of the rectangle.

Perimeter: \_\_\_\_\_





A square has sides of 9 inches.

What is its perimeter?

Perimeter: \_\_\_\_\_

## True or false?

A rectangle that is 10 m by 4 m has a perimeter of 28 m.

**A.** True

**B.** False

A triangle has sides of 5 cm, 6 cm, and 7 cm.

What is its perimeter?

- A.** 11 cm
- B.** 18 cm
- C.** 30 cm
- D.** 210 cm

A rectangle is 12 ft long and 5 ft wide.

What is its perimeter?

Perimeter: \_\_\_\_\_

SET 64 • 6.G.A.3

# FORWARD 1

Lengths between points with rational coordinates • Grade 7 • The Number System • 7.NS.A.1.c

A side of a polygon goes from  $(-2.5, 3)$  to  $(4, 3)$ .

How long is the side?

Length: \_\_\_\_\_

A rectangle has vertices at  $(-1.5, 2)$ ,  $(3.5, 2)$ ,  $(3.5, -1)$ , and  $(-1.5, -1)$ .

What is its perimeter?

Perimeter: \_\_\_\_\_

How long is the segment from  $(0, -3\frac{1}{2})$  to  $(0, 1\frac{1}{4})$ ?

**A.**  $-4\frac{3}{4}$

**B.**  $2\frac{1}{4}$

**C.**  $4\frac{3}{4}$

**D.**  $5\frac{1}{4}$



A square has vertices at  $(-2.5, -2.5)$ ,  $(2.5, -2.5)$ ,  $(2.5, 2.5)$ , and  $(-2.5, 2.5)$ .

What is the length of one side?

Length: \_\_\_\_\_

## True or false?

The segment from  $(-4.2, 1)$  to  $(1.8, 1)$  is 6 units long.

- A. True
- B. False

SET 64 • 6.G.A.3

# FORWARD 2

Find slanted side lengths with the Pythagorean Theorem • Grade 8 • Geometry  
• 8.G.B.8

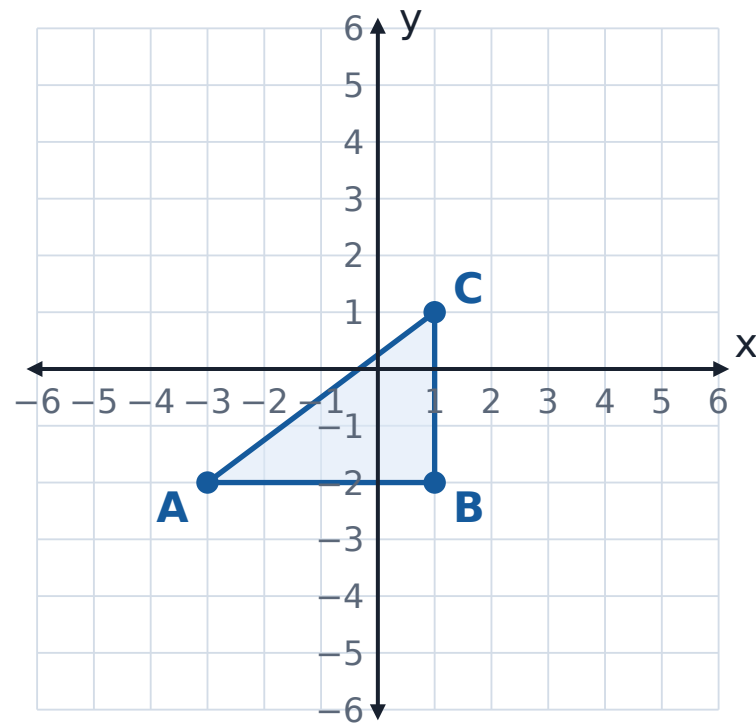
A triangle has vertices at  $(0, 0)$ ,  $(6, 0)$ , and  $(6, 8)$ .

How long is the slanted side?

Length: \_\_\_\_\_

Find the length of side AC.

AC = \_\_\_\_\_



A parallelogram has a side from  $(1, 1)$  to  $(4, 5)$ .

How long is that side?

- A.** 3
- B.** 5
- C.** 7
- D.** 25

Find the perimeter of the triangle with vertices  $(0, 0)$ ,  $(5, 0)$ , and  $(0, 12)$ .

Perimeter: \_\_\_\_\_

## True or false?

The side from  $(-2, 3)$  to  $(4, -5)$  is 10 units long.

**A.** True

**B.** False



# Represent three-dimensional figures using nets

|                   |                                             |                |
|-------------------|---------------------------------------------|----------------|
| <b>MAIN</b>       | Grade 6 main question                       | <b>6.G.A.4</b> |
| <b>BACKWARD 1</b> | Faces of three-dimensional shapes           | <b>2.G.A.1</b> |
| <b>BACKWARD 2</b> | Shapes of faces                             | <b>3.G.A.1</b> |
| <b>FORWARD 1</b>  | Cross sections of three-dimensional figures | <b>7.G.A.3</b> |
| <b>FORWARD 2</b>  | Volume of a cylinder described by its net   | <b>8.G.C.9</b> |

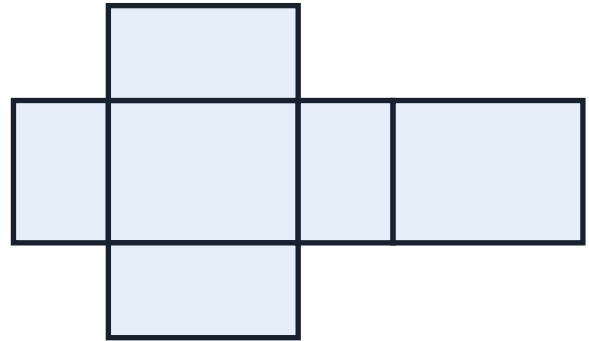
SET 65 • 6.G.A.4

# MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.4

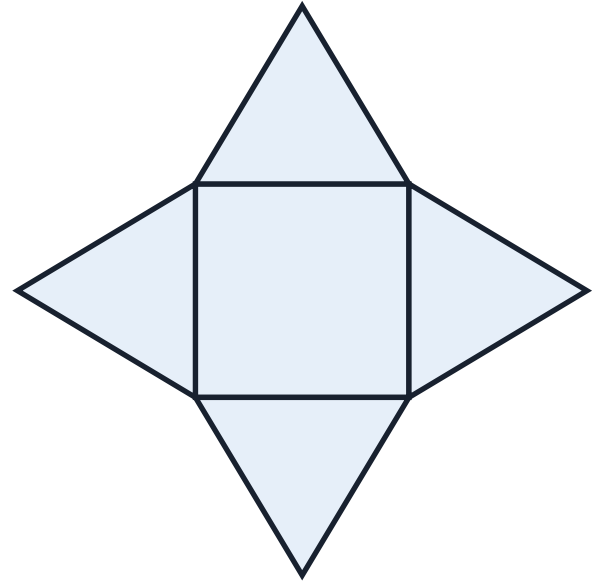
What three-dimensional figure does the net form?

- A.** Triangular prism
- B.** Rectangular prism
- C.** Square pyramid
- D.** Cube



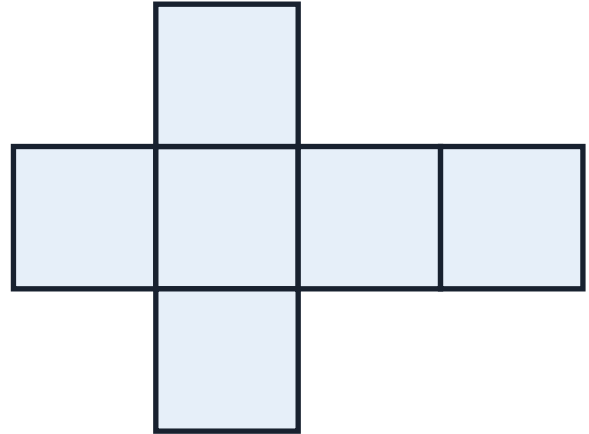
What three-dimensional figure does the net form?

- A.** Cube
- B.** Triangular pyramid
- C.** Square pyramid
- D.** Triangular prism



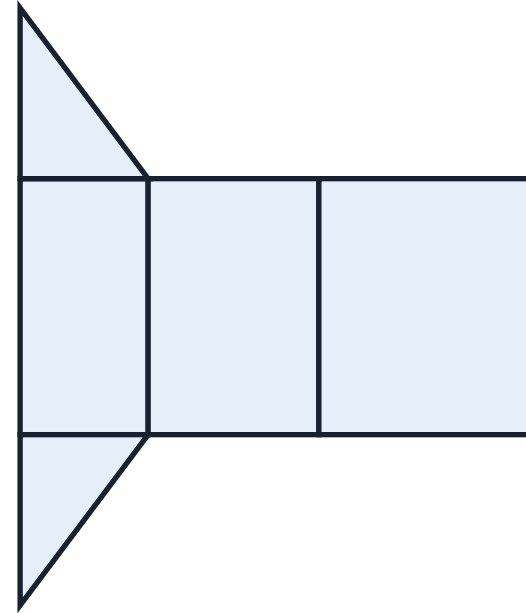
What three-dimensional figure does the net form?

- A.** Rectangular prism
- B.** Triangular prism
- C.** Cube
- D.** Square pyramid



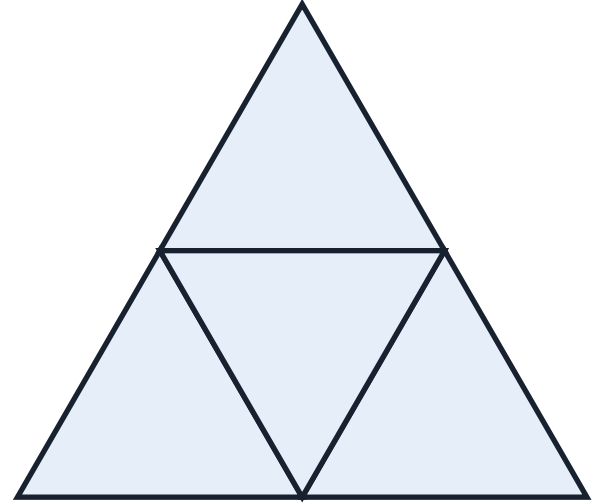
What three-dimensional figure does the net form?

- A.** Triangular prism
- B.** Triangular pyramid
- C.** Square pyramid
- D.** Rectangular prism



What three-dimensional figure does the net form?

- A.** Cube
- B.** Square pyramid
- C.** Triangular pyramid
- D.** Triangular prism



SET 65 • 6.G.A.4

# BACKWARD 1

Faces of three-dimensional shapes • Grade 2 • Geometry • 2.G.A.1